

Ising Machine Based on Electrically Coupled Spin Hall Nano-Oscillators

by

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Abstract

The Ising machine is an unconventional computing architecture that can solve NP-hard combinatorial optimization problems more efficiently than traditional von Neumann computing architectures. The spin Hall nano-oscillator has potential as a building block for a high-speed, low-power Ising machine based on its GHz operating frequency, sub-micron dimensions, and high degree of tunability. We develop an analytical framework describing how the dynamics of an electrically coupled array of spin Hall oscillators can be mapped to the Ising Hamiltonian based on the device characteristics. Our analytical model is integrated into a lightweight and versatile Verilog-A device that is used to model the nonlinear spin Hall oscillator's phase dynamics in SPICE-based circuit simulators. Finally, by integrating this device model with off-the-shelf electronic amplifier models, we analyze the Ising machine performance at the circuit level considering phase noise and scalability of the coupled network. The physics-based analytical models and quantitative tools presented in this work will enable future experimental realization of an electrically coupled spin Hall oscillator-based Ising machine operating with a high degree of time, space, and energy efficiency.

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Chapter 1

Introduction

As Moore’s Law scaling comes to an end, innovative devices and architectures are becoming necessary to enable further improvements in computing efficiency. One such alternative computing paradigm is the Ising machine, a non-von Neumann computing architecture that has been proposed to solve combinatorial optimization problems efficiently. Combinatorial optimization problems are ubiquitous in real-world applications such as VLSI design, operations research, computer networking, and industries such as glass and steel-cutting [63]. However, most problems of this class remain unsolvable in their full form on existing digital hardware due to their computational complexity. On the other hand, the Ising machine is naturally suited to solve these problems by using specialized hardware that follows a dynamic energy minimization process equivalent to finding the optimal solution of the problem.

Ising machines have been realized based on quantum annealing [15], optical parametric oscillators [27], and most recently electronic LC [75, 14] and phase-transition nano oscillators [20]. As opposed to quantum and optical implementations, the electronic oscillator-based Ising machines can be operated on chip-scale and at room temperature, making this approach most promising for realizing a scalable Ising machine. Experimental and theoretical demonstrations of the Ising machine using kHz-MHz electronic oscillators have proven the ability of the coupled oscillator network to achieve good solution accuracies on millisecond time scales. However, the Ising machine dynamics and solution times can be significantly sped up using oscillators

operating at higher frequencies.

Spin torque and spin Hall oscillators operating in the GHz frequency range are an emerging technology of interest for high-speed computing applications. These highly nonlinear oscillators have potential to realize an Ising machine with nanosecond-timescale dynamics. Previous works addressing spin torque oscillators for the Ising machine application have demonstrated this high-speed performance numerically based on a general coupling framework [4] and experimentally using a small network of magnetically coupled oscillators [28]. In our approach, we focus in detail on the electrical coupling dynamics of spin torque oscillators because we believe this coupling mechanism provides the most direct path toward an experimentally realizable and scalable Ising machine. Our proposed Ising machine uses a three-terminal spin Hall oscillator that provides additional tunability and ease of integration with existing Si technologies [47]. This oscillator topology in particular is an attractive technology for developing a fast, energy-efficient, and highly tunable Ising machine.

In this thesis, we evaluate the feasibility and performance of using electrically coupled spin Hall oscillators for the Ising machine through detailed physics- and circuit-level simulations. First, we study the dynamics of a spin torque oscillator coupled to an external radio-frequency signal and introduce an analytical model that can quantify the coupling efficiency between a network of electrically interconnected oscillators. This model, which is based on the fundamental device physics and characteristics, can be directly mapped to the Ising model and used to represent combinatorial optimization problems. The analytical model for electrical coupling dynamics is then integrated into a physics-based oscillator macromodel that is used in simulation with off-the-shelf electronic components to analyze the Ising machine at the circuit level. Finally, we study the statistical performance of the spin Hall oscillator Ising machine considering phase noise and scalability of the coupled network. By providing analytical and quantitative tools for modeling an electrically coupled network of spin Hall oscillators, our work enables the future experimental realization of a fast, energy-efficient spintronic Ising machine.

1.1 Motivation and Related Work

1.1.1 Existing Combinatorial Optimization Solvers

Due to their value in real-world applications, there is a significant push in modern computing research to develop methods to solve combinatorial optimization problems. Innovative software and hardware systems have been proposed based on different paradigms, from physics-inspired algorithms running on CPUs, GPUs, or FPGAs to specialized Ising machine architectures using a variety of hardware building blocks. In this section, we give a brief overview of the existing combinatorial optimization solvers and highlight the strengths and weaknesses of each approach.

CPU/GPU Simulated Annealing

Simulated annealing is an algorithmic method for solving combinatorial optimization problems inspired by the annealing of solids [72]. When a solid is exposed to a sufficiently high temperature then cooled slowly, the atoms of the solid are expected to rearrange into lower and lower energy states as the temperature is ramped down. By the end temperature, the solid is expected to reach the minimum-energy ground state of the material lattice. By analogy, the solution of a combinatorial optimization problem can be found by simulating the annealing process to minimize the problem’s equivalent energy, with the minimum-energy state being the optimal solution.

Simulated annealing algorithms are relatively easy to implement and can model a wide array of combinatorial optimization problems. However, because these algorithms must run over many iterations to return favorable results, they are computationally expensive and limited by long runtimes [30]. To mitigate the runtime issue, recent approaches have employed higher-performing digital hardware such as GPUs and FPGAs [23, 81]. Though the hardware is faster, these approaches do not address the fundamental issue that digital hardware is not well-suited to solve combinatorial optimization problems. Therefore, the algorithmic approach is not suitable for solving combinatorial optimization problems up to a real-world scale.

Quantum Annealing

Quantum annealing is relatively well-known due to its commercial application in the D-Wave quantum annealing system. So far, the quantum annealer is the only Ising Hamiltonian solver available in a commercial product. The D-Wave system implements an Ising machine using superconducting qubits coupled by Josephson junctions operating at cryogenic temperatures (around 15 mK) [32]. The most obvious drawback of this method is the requirement of cryogenic cooling, as such cooling systems are large, expensive, and consume significant energy. Second, a common problem across quantum systems is the limited numerical precision due to a high level of noise.

The relatively small size and sparse connectivity of the D-Wave computer introduce another obstacle for solving larger combinatorial optimization problems. The largest D-Wave computer to date has 5640 qubits (D-Wave Advantage, 2020). While the number of qubits in the D-Wave system has increased significantly over the years, the system’s connectivity is still limited to the chimera architecture. Many problems mapped onto the D-Wave architecture must be represented as a quadratic unconstrained Boolean optimization problem with each logical variable being represented by multiple qubits [16]. For these problems, the number of variables that can be represented is strictly less than the number of qubits. To solve full-scale combinatorial optimization problems, an even larger coupled system of qubits would be necessary.

Optical Parametric Oscillator

Another class of Ising machines has been demonstrated up to 2000 nodes based on optical parametric oscillators [29]. These solvers take advantage of the similarities between the Ising Hamiltonian used to represent combinatorial optimization problems and the Hamiltonian describing interfering coherent optical states. The optical coherent Ising machine involves the generation of short laser pulses that are circulated in a fiber ring cavity up to 1 km long.

While optical implementations of the Ising machine have demonstrated high

accuracy and efficient solution times, they require long fiber optic cables and complex materials and laser systems [51]. These hardware systems are not practical for building dense oscillator networks that will be necessary as the Ising machine is scaled. In addition, the generation, interference, and detection of optical pulses is highly phase sensitive and susceptible to external perturbations that can easily degrade the performance.

Electronic Oscillator

The general theory of the oscillator-based Ising machine was first proposed by Wang *et al.* [75] (this is not to be confused with the optical parametric oscillator-based Ising machine, which operates on a different physical principle). This work demonstrated how an Ising machine can be built using *any* nonlinear oscillator regardless of the specific physical operating principle and represents a significant step toward the realization of a scalable Ising machine using chip-scale electronic oscillators.

Multiple experimental demonstrations of the electronic oscillator-based Ising machine have appeared since that use self-sustaining LC oscillators [76, 14] and emerging phase-transition nano-oscillators [21]. The solution times of the oscillator-based Ising machine have been predicted to scale more slowly with the size of the coupled array than GPU and optical approaches [14]. However, electronic oscillators mainly operate in the kHz-MHz frequency regime, so the solution times remain on the order of ms even for relatively small coupled arrays. The solution time of the oscillator-based Ising machine is expected to scale inversely with the oscillator frequency [14], so this approach could benefit significantly from using higher-frequency oscillators.

Nano-Magnetic Oscillator

Nano-magnetic oscillators operating in the GHz frequency range represent a promising technology for high-speed computing arrays and a possible alternative to electronic oscillators for the oscillator-based Ising machine. A previous work based on general analytical formulas describing the spin torque oscillator operation has demonstrated that the nonlinear spin torque oscillator has the requisite characteristics to realize an

ultrafast Ising machine [4].

Because spin torque oscillators are a relatively young technology, there are still questions as to how large arrays of spin torque oscillators can be synchronized to act as a coherent computing unit. So far, only a few electrically coupled spin torque oscillators have been successfully applied to computing tasks [57, 74]. For the Ising machine, one work has demonstrated a small array of spin Hall oscillators coupled via magnetic fields that can be theoretically extended to solve combinatorial optimization problems [28]. However, it is unclear how the magnetic coupling mechanism can be made tunable enough to solve general optimization problems, as changing the magnitude of the magnetodipolar interaction would generally involve physically moving the oscillators in space relative to each other.

We believe that the electrical coupling of magnetic oscillators represents the most realistic direction toward an experimentally realizable spin torque oscillator-based Ising machine. By employing the highly tunable three-terminal spin Hall oscillator topology, we hope that our approach could mitigate some of the common issues in synchronizing arrays of oscillators such as frequency spread. Our goal is to provide detailed analytical and quantitative tools to model the electrically coupled spin Hall oscillator Ising machine and enable the future experimental realization of a scalable GHz-frequency Ising machine.

1.2 Modeling Approach

This work uses a variety of simulation tools to model the spin Hall oscillator Ising machine at varying levels of abstraction from the device to system level. In this section, we outline the main simulation methods used and their roles in this work.

For detailed physics-based simulations, we use the open-source micromagnetic simulation code `mumax3` [73]. `Mumax3` is a GPU-accelerated micromagnetic simulation code that allows for efficient simulation of magnetic materials and devices at a sub-micron scale. In micromagnetic simulations, a material or device of interest is split into discretization cells that are each assumed to have a uniform magnetic moment.

Then, the magnetic dynamics of each cell are evaluated by a fundamental equation governing the dynamics of a magnetic system, the LLGS equation (see Section 3.1) based on the cell’s interactions with its neighbors and external driving forces. Because of the fine detail, micromagnetic simulations provide a relatively physically accurate method for probing the dynamics of the spin torque oscillator. Throughout this work, we will use micromagnetic simulations as a benchmark to validate our own analytical models.

Next, to define a custom device model of the spin torque oscillator, we use the analog circuit modeling language Verilog-A. Verilog-A is most commonly used for writing custom device models to be used in SPICE-based circuit simulators. Though the spin torque oscillator is not a conventional circuit element, Verilog-A allows for great flexibility in defining the equations governing the dynamics of the oscillator and can therefore be used to build a versatile physics-based model of the spin torque oscillator. We subsequently use this device model in HSPICE simulations, where we model the spin torque oscillator alongside conventional electronic components to propose a system-level implementation of the Ising machine. HSPICE is a commercial analog circuit simulation software that can efficiently simulate circuit dynamics, allowing us to analyze our Ising machine up to a large size of 256 oscillators.

The technical content of this thesis begins with an analytical description of the coupling dynamics between the spin torque oscillator and an external applied radio-frequency signal. Based on this theory, we develop the spin torque oscillator macromodel in Verilog-A. This macromodel is used to model the electrical coupling behavior of an array of spin torque oscillators in circuit-level simulations. At each level of abstraction, we validate our results against benchmarks calculated by micromagnetic simulations to ensure that our analytical approach remains close to physical reality.

1.3 Outline

This thesis is divided into four main parts. In Chapters 2-3, we provide theoretical background on the Ising machine and spin Hall oscillator. In Chapter 4, we derive

an analytical model that links the Ising machine and spin Hall oscillator theory, demonstrating exactly how the physical dynamics of an electrically coupled network of spin Hall oscillators map to the Ising model. In Chapter 5, we integrate our detailed analytical model into a spin Hall oscillator device macromodel and demonstrate the use of macromodel in circuit-level simulations to model electrically interconnected oscillators. In Chapter 6, we evaluate the statistical performance of the spin Hall oscillator Ising machine on various benchmarking problems.

Chapter 2

Ising Machine Theory

2.1 Statistical Mechanical Ising Model

The Ising machine is inspired by the Ising model in statistical physics, which describes domain formation in ferromagnets via interactions between atomic spins arranged in a lattice [42]. The Ising Hamiltonian is given by [31]

$$H = - \sum_{i,j,i \neq j}^N J_{ij} s_i s_j - \sum_{i=1}^N h_i s_i \quad (2.1)$$

where s_i are discrete variables representing spins, of which there are N . Spins may be oriented in the “up” or “down” configuration corresponding to binary values $s = \pm 1$. J_{ij} are exchange coupling coefficients between spins that describe both the magnitude and polarity of their interaction. The sum over h_i is the Zeeman energy due to an external applied magnetic field. Many combinatorial optimization problems can be modeled by a simplified form that excludes the Zeeman term

$$H = - \sum_{i,j,i \neq j}^N J_{ij} s_i s_j \quad (2.2)$$

The Ising Hamiltonian represents the total energy of the ferromagnetic system based on the spin configuration. Naturally, a physical system favorably settles to its minimum-energy configuration, called the ground state. If the ferromagnetic material

is not yet in the ground state, the spins s will reorient themselves based on the interaction terms J_{ij} to minimize H defined by Equation 2.1 or 2.2. A similar theory can be applied to solving combinatorial optimization problems, where the problem variables are “reoriented” in order to minimize the problem’s equivalent energy (find the optimal solution). In the following section, we show how the mapping between a combinatorial optimization problem and the Ising Hamiltonian can be done.

2.2 Mapping to Combinatorial Optimization Problems

Just as the Ising Hamiltonian describes the energy of an assembly of spins in a ferromagnet, we can define an analogous energy function describing a combinatorial optimization problem of interest. The spins s represent variables to optimize on, and the energy function is defined such that the global minimum value of H (the Ising Hamiltonian ground state) corresponds to a spin state encoding the optimal solution of the problem. In this way, the energy minimization dynamics of a physical system described by the Ising Hamiltonian will naturally lead to the optimal solution of a combinatorial optimization problem. This approach is applicable to combinatorial optimization problems of many forms. As shown by Lucas, all 21 problems in Karp’s well-known set of NP-complete problems can be mapped to an equivalent Ising Hamiltonian [48, 33].

As a simple illustrative example, we show the equivalent Ising Hamiltonian representation of the maximum cut graph partitioning problem (MAX-CUT) [33, 79]. This problem involves dividing an undirected weighted graph into two subgraphs such that the sum of weights of edges connecting the two subgraphs is maximized. A small undirected weighted graph is shown in Figure 2-1. The graph nodes are represented by spins and the spin value $s = \pm 1$ represents which of the two subgraphs the node belongs to: either the spin-up ($s = +1$) or spin-down ($s = -1$) subgraph. The coupling coefficients J_{ij} are set as the negative of the edge weights w_{ij} linking nodes i and j .

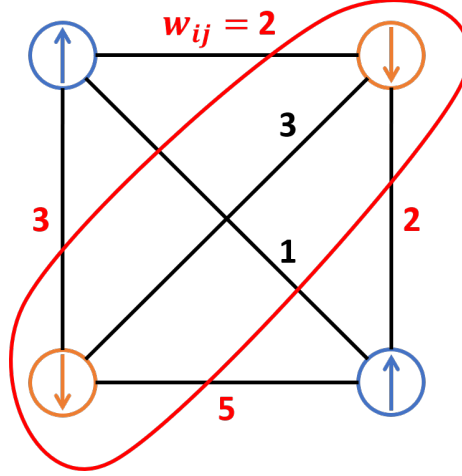


Figure 2-1: 4-node undirected weighted graph on which the MAX-CUT problem is modeled. The edge weights w_{ij} are labeled. The nodes are split into two subgraphs (spin-up and spin-down, denoted by arrows on the nodes). The curve partitioning the graph into two subgraphs is drawn and the weights of the cut edges are highlighted in red (the twice-cut edges do not count as being cut).

Therefore, the equivalent MAX-CUT Ising Hamiltonian is given by

$$H_{cut} = \sum_{i,j,i \neq j}^N w_{ij} s_i s_j \quad (2.3)$$

The ground state of Equation 2.3 (maximum cut) is achieved when the edges linking two opposite-spin nodes (in separate subgraphs, such that $s_i s_j = -1$) have the largest positive weights and edges linking two same-spin nodes (in the same subgraph, such that $s_i s_j = +1$) have the smallest weights. By inspection, one can see that the graph partition in Figure 2-1 corresponds to the ground state of Equation 2.3.

2.3 Oscillator-Based Ising Machine

Using a simple example, we have shown how a mapping can be made between a combinatorial optimization problem and the Ising Hamiltonian. In this section, we show how the Ising Hamiltonian can be emulated by the phase dynamics of a coupled network of nonlinear oscillators. A small coupled network of oscillators is shown schematically in Figure 2-2.

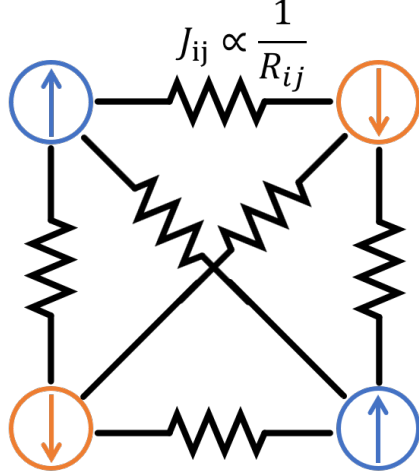


Figure 2-2: Schematic of the electrically coupled oscillator-based Ising machine. This 4-oscillator network can be used to model the MAX-CUT problem shown in Figure 2-1 by setting the appropriate coupling coefficients (weighting conductances) $J_{ij} = -w_{ij} \propto 1/R_{ij}$. Negative weighting conductances are achieved by inverting the signal between oscillator i and j .

When a radio-frequency signal is applied to a nonlinear oscillator, the oscillator's frequency and/or phase can match that of the external signal by a process known as injection locking. Injection locking is described by the well-known Adler's equation [2], shown below for a coupled network of N oscillators [41]

$$\frac{d\phi_i}{dt} = \omega_i - \omega^* - \omega_i \sum_j^N A_{ij} \sin(\phi_i(t) - \phi_j(t)) \quad (2.4)$$

where ω_i and ϕ_i are the frequency and phase of oscillator i , ω^* is the mean frequency of the oscillators in the coupled network, ϕ_j is the phase of oscillator j , and A_{ij} is the injection locking strength between coupled oscillators i and j . In an electrically coupled network of oscillators, the injection locking strength A_{ij} can be set by the conductances $G_{ij} = 1/R_{ij}$ linking oscillators i and j , as we will derive explicitly in Section 4.2.

The collective phase dynamics of the oscillators minimize the array's interaction

potential H as derived in [41]

$$H = - \sum_{i,j,i \neq j}^N K_{ij} \cos(\phi_i - \phi_j) \quad (2.5)$$

that is related to the Ising Hamiltonian in Equation 2.2 by coupling coefficients $K_{ij} = J_{ij}$ and relative oscillator phases $(\phi_i - \phi_j) = [0, \pi] \mapsto s_i s_j = [+1, -1]$. Intuitively, the synchronization of each coupled oscillator's phase to its injected signal corresponds to the minimization of that oscillator's contribution to the interaction potential, where in the ground state $\frac{d\phi_i}{dt} = -\frac{\partial H}{\partial \phi_i} = 0$ (neglecting local minima or saddle points) [59].

To map between the continuous-time phase dynamics of the oscillator network and the discrete Ising Hamiltonian, a global second harmonic signal is applied to the oscillator network. In Adler's equation, this introduces a second term [75]

$$\frac{d\phi_i}{dt} = \omega_i - \omega^* - \omega_i \sum_j^N A_{ij} \sin(\phi_i(t) - \phi_j(t)) - \omega_i A_s \sin(2\phi_i(t)) \quad (2.6)$$

with the injection locking strength of the second harmonic signal represented by A_s . This introduces a second term in the interaction potential [75]

$$H = - \sum_{i,j,i \neq j}^N K_{ij} \cos(\phi_i - \phi_j) - \sum_i^N K_s \cos(2\phi_i) \quad (2.7)$$

with second harmonic injection locking strength K_s . From the interaction potential, we see that the application of a second harmonic injection locking signal introduces additional energy minima at $\phi_i = [0, \pi]$, encouraging the phases to binarize to these values corresponding to spins $s_i = [+1, -1]$ in the Ising model. The introduction of the second harmonic signal does not change the relative spacings of the energy levels represented by the Ising Hamiltonian at the discrete points, but rather applies a constant energy offset that does not affect the overall ground state search [75].

Chapter 3

Spin Hall Nano-Oscillator

This chapter is dedicated to the theoretical background and micromagnetic modeling of the spin Hall nano-oscillator. Spin torque and spin Hall nano-oscillators are emerging spintronic technologies that are being explored for unconventional computing applications. In general, the spin-based oscillator consists of a nanomagnet excited by spin-polarized current, leading to sustained auto-oscillations of the magnetic moments. These oscillators are highly nonlinear and tunable based on multiple properties including the geometry, material, and excitation mechanism. Spin torque oscillators operate at high frequencies from 100's of MHz to well into the GHz range with sub-micron dimensions, making them attractive elements for use in dense, high-speed computing arrays.

3.1 Theory of Operation

In this section, we briefly summarize the physical mechanisms underlying the spin torque and spin Hall nano-oscillator operation and the common oscillator topologies. The analytical equation governing the oscillator's equilibrium oscillation state will be introduced next in Section 3.2.

A nanomagnet excited by spin-polarized current can be used to realize a self-sustaining nonlinear oscillator, called a spin torque or spin Hall nano-oscillator depending on the exact excitation mechanism. Consider a magnetic multilayer consisting of

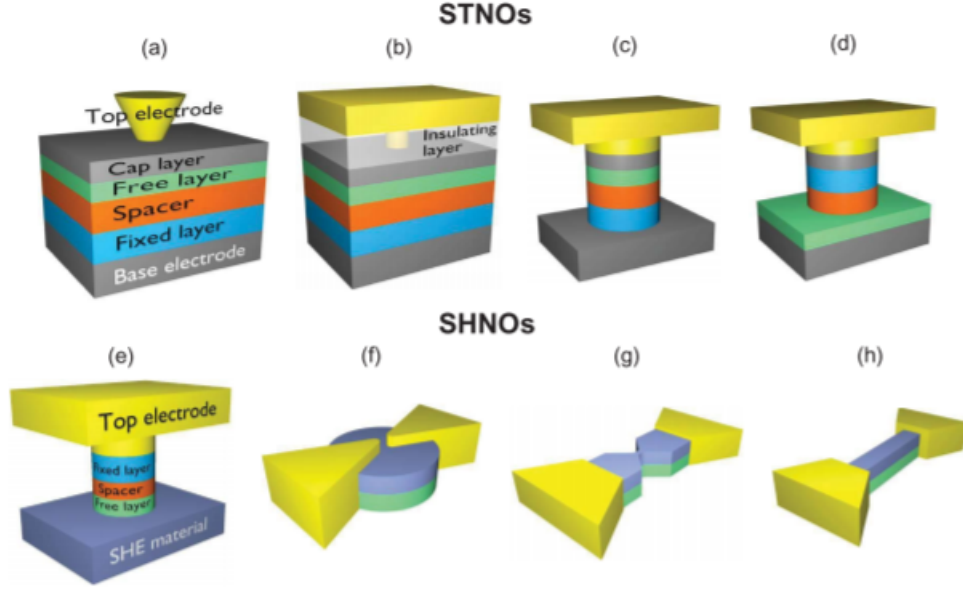


Figure 3-1: Schematics of common STNO and SHNO device architectures, including (a) point contact, (b) nano-contact, (c) nano-pillar, (d) hybrid, (e) nano-pillar, (f) nano-gap, (g) nano-constriction, (h) nano-wire. This figure was originally published in [11].

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two ferromagnetic layers (free, fixed layers) separated by an insulating tunnel barrier (spacer) shown in Figure 3-1a-e. The magnetic moments in the fixed layer are pinned along a specific direction by exchange bias with a neighboring magnetic layer or by using a material with relatively large anisotropy or thickness. On the other hand, the moments in the free layer are free to move and rotate under the influence of applied fields and spin currents.

To excite auto-oscillations of the free layer magnetization, a magnetic field is first applied to define the static equilibrium position of the free layer moments. When a sufficiently high spin current excites the magnetic free layer moments, the moments enter a coherent precession about their static equilibrium position [11]. In a spin torque oscillator, this spin current is supplied by passing an unpolarized charge through the magnetic fixed layer to become spin polarized. In a spin Hall oscillator, the charge current is applied to a heavy metal (SHE material) with strong spin-orbit interaction such that a spin polarized current is generated by the spin Hall effect (SHE) [1]. The

spin current flows along the normal direction of the heavy metal layer with the spin polarization direction transverse to both the charge and spin current flow directions [47, 66].

The readout of the oscillating magnetization is based on the tunnel magnetoresistance effect (TMR) [82]. TMR is based on the quantum mechanical tunneling of electrons through the insulating spacer layer where the tunneling probability depends on the relative orientation of the free and fixed layer magnetic moments. The electron tunneling probability determines the resistance experienced by a charge current flowing through the nanopillar, enabling electrical readout of the oscillating moment by a radio-frequency oscillating magnetoresistance. In spin torque nano-oscillators, the charge current used to excite auto-oscillations also acts as the readout current to generate a microwave voltage via the TMR effect. In the 3-terminal spin Hall nano-oscillator [47], however, the charge current in the heavy metal layer flows transverse to the nanopillar rather than through it, so a dedicated readout voltage is applied to the top electrode (Figure 3-1e). In Section 3.1.1, we will describe how this separation of the excitation and readout mechanisms enables further tunability of the spin Hall nano-oscillator device.

The additional spin Hall nano-oscillator topologies shown in Figure 3-1f-h are not discussed in this thesis, but further details can be found in [11].

3.1.1 Three-Terminal Spin Hall Oscillator Topology

The three-terminal spin Hall nano-oscillator is of particular interest in computing based on its ease of integration with Si-based electronics and enhanced tunability resulting from multiple excitation mechanisms [47]. The spin Hall oscillator is also more efficient at exciting free layer auto-oscillations in certain geometries [46], meaning that a spin Hall oscillator-based Ising machine could potentially achieve a lower operating power than a spin torque oscillator-based version.

Besides the charge current flowing through the spin Hall layer to excite auto-oscillations of the magnetic free layer, the MTJ terminal of the spin Hall oscillator provides a secondary excitation mechanism and dynamic tuning knob. The primary

purpose of a DC voltage applied to the MTJ terminal is to enable electrical readout of the magnetization by TMR. This charge current is also spin-polarized by the fixed layer and can act with or against the spin current from the heavy metal layer to contribute to the auto-oscillatory dynamics of the free layer.

The MTJ voltage can also modulate the effective magnetic field experienced by the free layer by inducing an electric field across the insulating oxide tunnel barrier. The electric field leads to voltage-induced perpendicular magnetic anisotropy (PMA) in the free layer [52]. PMA reduces the effective out-of-plane demagnetizing field, allowing the free layer magnetic moments to move more freely in the out-of-plane direction. The opposite effect can also be realized where the MTJ voltage induces “negative” PMA to strengthen the out-of-plane demagnetizing field. The strong nonlinearity of the spin Hall oscillator means that the frequency, linewidth, and power of the auto-oscillations can all be simultaneously modulated by the spin Hall and MTJ excitation mechanisms, resulting in a dynamically tunable auto-oscillator with favorable characteristics to be integrated in a large-scale computing array [61].

As briefly mentioned in Section 1.1.1, frequency spread can present a significant obstacle in synchronizing large coupled arrays of spin torque oscillators for computing applications. Using the tunable spin Hall oscillator, this issue can be mitigated by introducing a voltage-controlled oscillator (VCO) calibration scheme [83]. VCO calibration can be done by a one-time post-fabrication step where the DC voltages applied to the spin Hall layer and MTJ terminal are tuned within a finite range to bring all the oscillators in the array to a similar frequency. Based on the strong nonlinearity of the spin Hall nano-oscillator, VCO calibration can likely be used to achieve a close frequency match between the devices in a coupled array.

Of all the oscillator topologies represented in Figure 3-1a-e, the three-terminal spin Hall oscillator is most likely to be successfully fabricated and integrated into an electrically coupled Ising machine. First, excitation by the spin Hall effect does not require the charge current from the bottom electrode to flow perpendicularly upward through the nanopillar, so it is more simple to implement in an integrated circuit than a spin torque oscillator. The base heavy metal layer is also easily grown on Si

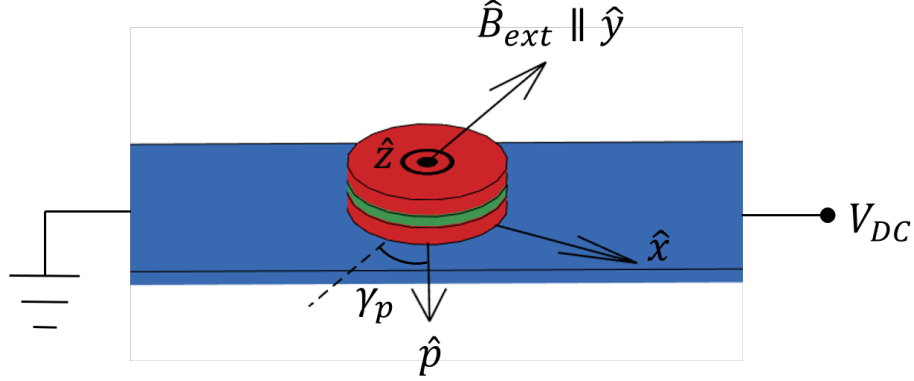


Figure 3-2: Axis definition of the 3-terminal spin Hall nano-oscillator used in micro-magnetic simulations. The MTJ stack consists of a CoFeB free layer (bottom red), MgO spacer layer (green), and CoFeB fixed layer (upper red) assumed to be polarized along $+\hat{x}$. The MTJ dimensions are not drawn to scale.

substrates that are used in conventional electronics. Based on its ease of integration with existing technologies, potential for low-power operation, and high degree of tunability, we choose the three-terminal spin Hall oscillator as the main building block of our oscillator-based Ising machine.

3.2 Equilibrium Auto-Oscillation

We describe the analytical equation governing the equilibrium magnetization precession of the in-plane magnetized spin torque oscillator shown in Figure 3-2. The macrospin Landau-Lifshitz-Gilbert-Slonczewski (LLGS) equation is used to describe the dynamics of the free layer magnetization under applied fields and spin currents [44, 24, 62]

$$\frac{d\vec{m}}{dt} = -\frac{\gamma_e}{1+\alpha^2} \left[\vec{m} \times \vec{B}_{eff} + \alpha (\vec{m} \times (\vec{m} \times \vec{B}_{eff})) + B_s (\vec{m} \times (\vec{m} \times \hat{p})) - \alpha B_s (\vec{m} \times \hat{p}) \right] \quad (3.1)$$

where γ_e is the electron gyromagnetic ratio, α is the Gilbert damping parameter, and $\vec{m}$ is the normalized free layer magnetization. Based on the device shown in Figure

3-2, the effective magnetic field is given by

$$\vec{B}_{eff} = \begin{pmatrix} -B_d m_x \\ B_{ext} - B_d m_y \\ (B_{PMA} - 4\pi M_s) m_z \end{pmatrix} \quad (3.2)$$

including contributions from the applied field B_{ext} , perpendicular magnetic anisotropy field B_{PMA} resulting from the interfacial and MTJ voltage-controlled magnetic anisotropy, in-plane demagnetizing field B_d based on the device geometry, and out-of-plane demagnetizing field $4\pi M_s$.

The spin torque excitation term is

$$B_s \hat{p} = \frac{\sigma V_{dc}}{R_{HM}} \begin{pmatrix} \sin(\gamma_p) \\ -\cos(\gamma_p) \\ 0 \end{pmatrix} \quad (3.3)$$

with $\sigma = \hbar\theta_{SH}/(2eM_s A t_{FL})$ where θ_{SH} is the spin Hall angle, M_s is the free layer saturation magnetization, A is the cross-sectional area of the heavy metal surface through which charge current is applied, t_{FL} is the magnetic free layer thickness, and R_{HM} is the total resistance of the heavy metal strip.

To excite persistent auto-oscillations of the free layer magnetization, the dissipation due to damping must be compensated by the energy supplied by spin torque over each cycle. The work done spin torque W_s and dissipation by damping W_α are given by [64, 65]

$$\begin{aligned} \frac{dE}{dt} &= -\frac{M_s}{\mu_0} \vec{B}_{eff} \cdot \frac{d\vec{m}}{dt} \\ &= -\frac{\gamma_e \alpha M_s}{\mu_0} [(\vec{m} \cdot \vec{B}_{eff})^2 - \vec{B}_{eff}^2] \\ &\quad + \frac{\gamma_e M_s B_{s0}}{\mu_0} [(\vec{m} \cdot \vec{B}_{eff})(\vec{m} \cdot \vec{p}) - (\vec{p} \cdot \vec{B}_{eff})] \\ &= W_\alpha + W_s \end{aligned} \quad (3.4)$$

where W_α comes from the second term of Equation 4.2 and W_s comes from the third and fourth terms, where the third term is dominant in the case of low magnetic

damping α . When a spatial trajectory exists where W_s and W_α compensate each other over each cycle ($\oint (W_s + W_\alpha) dt = \frac{dE}{dt} = 0$), the free layer magnetization stably precesses about the applied magnetic field as described by the first term in Equation 4.2.

The strong nonlinearity of the spin Hall oscillator arises from the complex interactions between the effective field and driving forces acting on the oscillating moment as described by Equation 3.4. Not only does a change of the driving voltage V_{DC} lead to a change of the oscillation frequency and amplitude, but the amplitude and phase of the oscillator are also coupled to each other via the dependence of the demagnetizing fields on the orbit of oscillation (Equation 3.2) [40]. This leads to effects such as large injection locking strength [61] and phase noise enhancement [36] in spin Hall oscillators operating in highly nonlinear regimes that could have interesting implications in spin Hall oscillator-based computing systems.

3.3 Micromagnetic Simulation

In this section, we introduce the modeling of the three-terminal spin Hall oscillator device in micromagnetic simulations. We show the equilibrium free layer magnetization precession at the chosen operating point that will be important later in our analytical model of injection locking.

The general axis definitions of the simulated spin Hall oscillator device are given in Figure 3-2. An elliptical MTJ nanopillar is simulated with in-plane dimensions 120×200 nm². Only the CoFeB MTJ free layer (with thickness $f_{FL} = 1.5$ nm) is explicitly modeled in the micromagnetic simulations. The CoFeB material parameters are as follows: saturation magnetization $M_s = 800$ kA/m, exchange stiffness $A_{ex} = 13$ pJ/m, Gilbert damping $\alpha = 0.02$, perpendicular magnetic anisotropy constant $K_u = 180$ kJ/m³. The free layer is discretized into cells with dimension $5 \times 5 \times 1.5$ nm³ in x, y, and z respectively. A magnetic field $B_{ext} = 20$ mT is applied in-plane along the short axis of the MTJ such that the equilibrium position of the free layer magnetic moments is along this axis. Thermal fluctuations are neglected in our simulations.

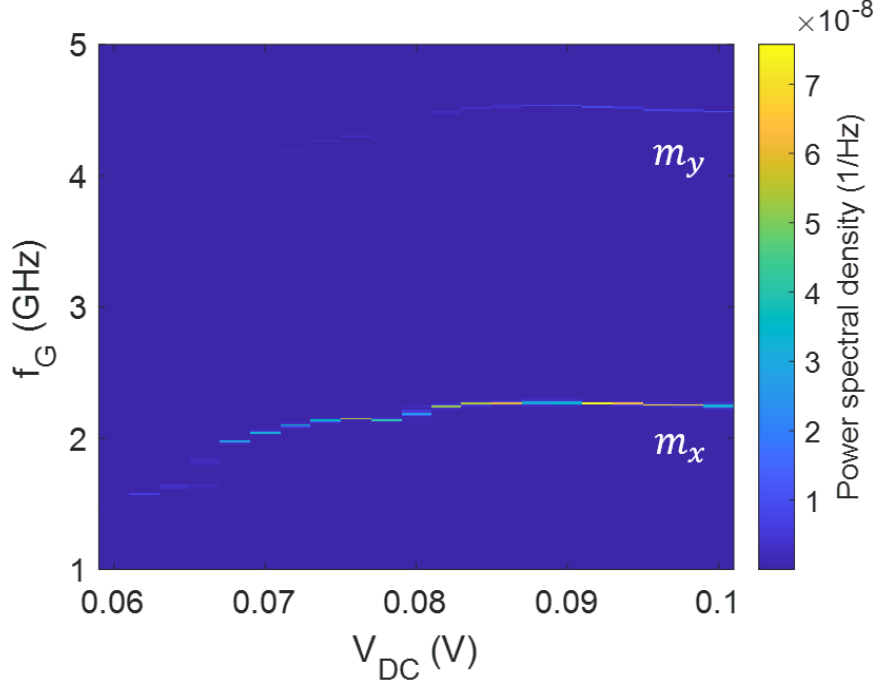


Figure 3-3: Free-running power spectral density of the in-plane normalized magnetization of the spin Hall oscillator as a function of V_{DC} .

The parameters of the spin injection to the CoFeB free layer are chosen assuming Pt as the heavy metal with the spin Hall angle $\theta_{SH} = 0.07$. The injected spin polarization direction $\hat{p}$ is shown in Figure 3-2 to be $\gamma_p = 30^\circ$ off the applied field direction $\hat{B}_{ext}$.

In Figure 3-3, we show the power spectral density of the normalized free-layer magnetization $\vec{m}$ as a function of V_{DC} . Only in-plane components m_x, m_y are shown. The measured threshold voltage to observe auto-oscillations is $V_{DC,th} = 60$ mV. We choose an operating point $V_{DC} = 88$ mV that is around 1.5 times the threshold value because of the large-angle oscillation in the $x-y$ plane (plotted spatially in Figure 3-4) and the relatively weak nonlinearity of the more strongly driven device that simplifies the injection locking dynamics [61].

One slice of the power spectral density plot at the operating point $V_{DC} = 88$ mV is shown in Figure 3-5 including m_x, m_y , and m_z components. We clearly see that the m_x and m_z components oscillate at the oscillator's fundamental frequency, while the m_y component oscillates at twice this frequency. The reason for this second harmonic component is due to the shape of the oscillation trajectory, shown in Figure 3-4. This

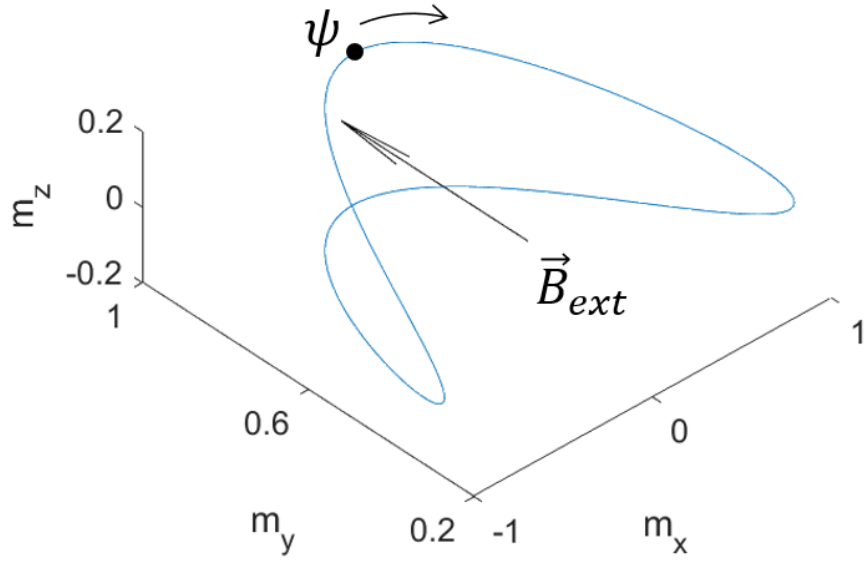


Figure 3-4: “Clamshell” shaped equilibrium precession $\vec{m}_{eq}$ of the in-plane magnetized spin Hall oscillator.

shape is classified as a “clamshell” orbit where the magnetization components transverse to the applied field oscillate at the fundamental frequency while the component parallel to the applied field oscillates at the second harmonic based on symmetry [38]. In Section 4.2, we will show that this clamshell-shaped precession enables strong injection locking between an external applied second harmonic signal and the second harmonic m_y component [71]. The strong second harmonic locking is important to realize the second harmonic phase binarization effect required to operate the oscillator-based Ising machine [75].

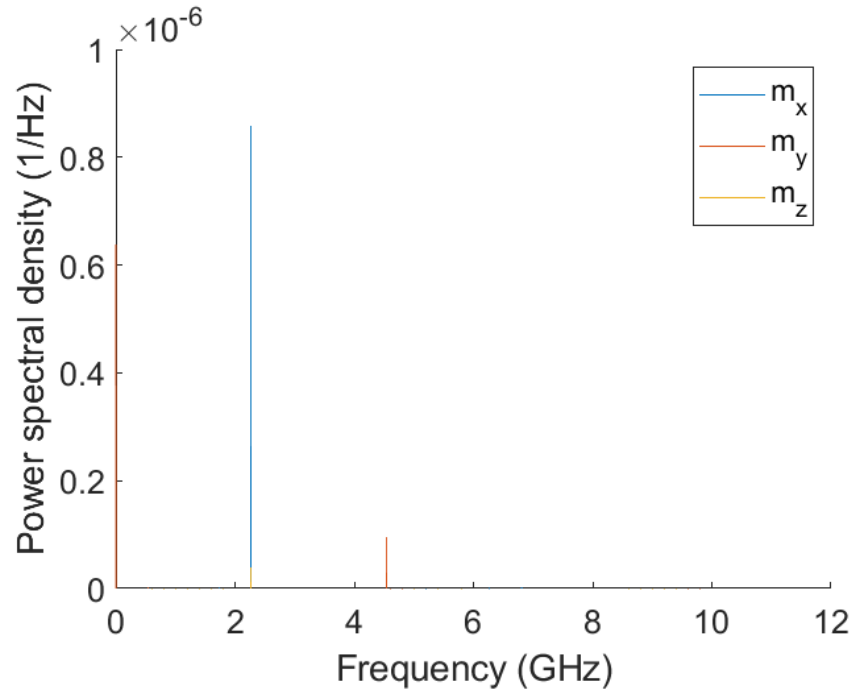


Figure 3-5: Power spectral density associated with the normalized magnetization components of the oscillator device simulated in micromagnetic simulations. In the weakly nonlinear oscillator with clamshell precession, the peaks for first and second harmonics are clearly visible while higher-harmonic components are not visible.

Chapter 4

Injection Locking Model of Spin Hall Nano-Oscillator

We derive an analytical model for the injection locking dynamics of the nonlinear spin Hall oscillator that are central to the operation of the oscillator-based Ising machine. Injection locking is a nonlinear process by which the frequency and/or phase of an oscillator matches that of an external signal. We focus on phase locking dynamics in particular, where the oscillator’s phase synchronizes to that of a locking signal with a frequency close to the oscillator’s own frequency or one of its harmonics [45]. Multiple important properties of the Ising machine directly depend on the strength of the phase locking effect, including the time-to-solution [14], phase binarization, and robustness to device variability. Therefore, it is important that we establish a comprehensive analytical formula that will allow us to predict the injection locking strength based on the spin Hall oscillator’s properties.

Previous works have directly derived relationships between the spin torque oscillator properties and injection locking behavior represented by Adler’s equation [61, 60]. However, these models rely on the assumption that the oscillation orbit is small and the oscillating moment does not deviate far from its static equilibrium position, limiting this approach to a very small subset of the oscillator’s wide range of operating points. The strength of the injection locking effect depends strongly on the shape of the equilibrium oscillation orbit, but the explicit relation has not yet been analytically

derived in the literature.

On the same note, due to the lack of a complete injection locking model, we cannot analytically predict the strength of the parametric locking effect whereby the oscillator locks to a signal at a multiple of the free-running frequency [71, 45]. Predicting the parametric locking strength of the spin Hall oscillator is important for determining an appropriate amplitude for the second harmonic injection locking signal used in the oscillator-based Ising machine. In addition, introducing an analytical model of the parametric locking effect will help us to gain more insight into the interesting physics of the nonlinear spin Hall oscillator.

Our general modeling approach is based on the impulse sensitivity function (ISF) describing the oscillator's phase and amplitude response to perturbation by an external signal [26]. If an impulse perturbation is applied to a nonlinear oscillator at time τ , the phase response takes the form

$$h_\phi(t, \tau) = \Gamma(\tau)u(t - \tau) \quad (4.1)$$

where $u(t - \tau)$ is a unit step and the coefficient $\Gamma(\tau)$ is the ISF [55]. The phase shift represents a translation along the oscillator's equilibrium trajectory and persists after the perturbation is removed. On the other hand, the amplitude response represents a deviation from the equilibrium trajectory that will decay in time due to the restoring force of the nonlinear oscillator to its steady-state trajectory [19]. Because we are most interested in the phase dynamics for analyzing injection locking, we neglect amplitude deviations [19, 50, 55, 39, 26, 7]. This is a reasonable approximation for a weakly nonlinear spin torque oscillator in which the phase and amplitude fluctuations are not strongly coupled [36, 40] or for a more strongly nonlinear oscillator perturbed by a weak signal.

The ISF is derived based on a linearization of the phase response about the oscillator's free-running trajectory. While the small-signal phase response is treated linearly, the ISF includes information about the oscillator's nonlinearity through its dependence on the oscillation trajectory. Previous works have calculated the ISF

through various approaches, such as nonlinear perturbation theory [19, 50], direct measurement of the oscillator's phase response to impulse perturbations [55, 39], or estimation of the phase shifts by projecting the perturbation along the equilibrium oscillation trajectory [26, 7]. We use the third method because it allows us to construct an intuitive analytical expression for the spin torque oscillator ISF.

In this chapter, we first discuss how radio-frequency dynamics are separated from the DC equilibrium magnetization precession in the LLGS equation. Then, we show how the ISF can be derived analytically from one period of the equilibrium oscillation trajectory. We focus on the ISF for first and second harmonic injection locking of an oscillator with large-angle precession, but the present approach can be generalized to a spin torque oscillator with any arbitrary precession orbit. Finally, we calculate Adler's equation based on the ISF and demonstrate agreement of the predicted injection locking behavior with micromagnetic simulations. Because Adler's equation is directly comparable to the Ising model as shown in Section 2.3, we explicitly show a mapping from the oscillator properties to an equivalent Ising Hamiltonian represented by an electrically coupled network of spin torque oscillators.

4.1 Inclusion of Radio-Frequency Spin Torques in Magnetic Dynamics

We build upon the analysis of the equilibrium magnetization precession in Section 3.2 to show how radio-frequency torques at the first and second harmonics are considered in the dynamical LLGS equation. Assuming low magnetic damping, the LLGS equation (Equation 3.1) can be approximated by

$$\begin{aligned}\frac{d\vec{m}}{dt} &= -\gamma_e [\vec{m} \times \vec{B}_{eff} + \alpha(\vec{m} \times (\vec{m} \times \vec{B}_{eff})) + (B_{s0} + b_{s1f} + b_{s2f})(\vec{m} \times (\vec{m} \times \hat{p}))] \\ &= \vec{\tau}_{eq} + \vec{\tau}_{1f} + \vec{\tau}_{2f}\end{aligned}\tag{4.2}$$

where we have split the spin torque field B_s into DC, first, and second harmonic components based on the different voltage signals applied to the heavy metal

$$\begin{aligned} B_s \hat{p} &= (B_{s0} + b_{s1f} + b_{s2f}) \hat{p} \\ &= \frac{\sigma}{R_{HM}} (V_{dc} + V_{1f}(t) + V_{2f}(t)) \begin{pmatrix} \sin(\gamma_p) \\ -\cos(\gamma_p) \\ 0 \end{pmatrix} \end{aligned} \quad (4.3)$$

We assume the radio-frequency (rf) spin torque fields b_{s1f} , b_{s2f} are relatively weak compared to the DC component B_{s0} . Therefore, the rf torques do not modulate the equilibrium magnetization precession based on $\vec{\tau}_{eq}$ evaluated in Section 3.2. Instead, these rf torques $\vec{\tau}_{1f}$, $\vec{\tau}_{2f}$ are used to calculate the ISFs for first and second harmonic injection locking in Section 4.2.

Before deriving the ISF, we must know the shape of the equilibrium magnetization precession. To leading order, the equilibrium precession simulated in Section 3.3 and plotted in 3-4 can be approximated by

$$\vec{m}_{eq} = \begin{pmatrix} m_x \sin(\psi) \\ m_{y0} + m_y \cos(2\psi) \\ m_z \cos(\psi) \end{pmatrix} \quad (4.4)$$

where we define the total oscillator phase $\psi = \omega_g t + \phi$. The leading-order approximation is valid for this weakly nonlinear oscillator as higher-harmonic peaks are not visible in the power spectral density plotted in Figure 3-5. To model higher-order injection locking effects in a more strongly nonlinear oscillator, one may choose to include more terms in Equation 4.4 describing the nonlinear harmonic generation of the spin torque oscillator [71].

4.2 Impulse Sensitivity Function for Injection Locking

We calculate the ISF of the spin torque oscillator based on a linearization of the phase dynamics about the steady-state precession trajectory. Our general expression is used to predict the injection locking sensitivity to rf spin torques b_{s1f}, b_{s2f} at the first and second harmonics in Equation 4.2. Our analysis begins with finding a closed-form expression for the ISF by calculating the projection of the perturbing force along the oscillator trajectory as demonstrated by Hajimiri *et al.* [26] Appendix B.

We consider a spin torque oscillator with equilibrium magnetization precession given by $\vec{m}_{eq}(t)$. Under an impulse perturbation $P(t) = \delta(t - t_0)$, we expect a response of the oscillator magnetization $\vec{m}(t_0^+) = \vec{m}_{eq}(t_0) + \Delta\vec{m}(t)$. Because we neglect amplitude deviations, we will only consider the component of $\Delta\vec{m}(t)$ along the equilibrium precession trajectory $\dot{\vec{m}}_{eq}$, representing a phase shift. The displacement l along the precession trajectory is given by $\Delta\vec{m}(t)$ dotted with a unit vector in the trajectory direction

$$l = \Delta\vec{m}(t) \cdot \frac{\dot{\vec{m}}_{eq}}{|\dot{\vec{m}}_{eq}|} \quad (4.5)$$

We then divide by the precession “speed” $|\dot{\vec{m}}_{eq}|$ to find the equivalent phase shift in units of time (denoted by the subscript t)

$$\Delta\phi_t = \Delta\vec{m}(t) \cdot \frac{\dot{\vec{m}}_{eq}}{|\dot{\vec{m}}_{eq}|^2} \quad (4.6)$$

By Equation 4.2, we can change the right side to an integral over the perturbing torque $\Delta\vec{\tau}(t)$

$$\begin{aligned} \Delta\phi_t &= \int_0^\infty \Delta\vec{\tau}(t)\delta(t - t_0) \cdot \frac{\dot{\vec{m}}_{eq}(t)}{|\dot{\vec{m}}_{eq}(t)|^2} dt \\ &= \Delta\vec{\tau}(t_0)u(t - t_0) \cdot \frac{\dot{\vec{m}}_{eq}(t_0)}{|\dot{\vec{m}}_{eq}(t_0)|^2} \end{aligned} \quad (4.7)$$

By considering this phase shift as the phase impulse response in Equation 4.1, we see

that the general expression for the ISF is

$$\Gamma(t) = \frac{\Delta \vec{\tau}(t)}{P} \cdot \frac{\dot{\vec{m}}_{eq}(t)}{|\dot{\vec{m}}_{eq}(t)|^2} \quad (4.8)$$

where we divide by the perturbation amplitude P in the case of a non-unit-amplitude perturbation. The impulse sensitivity function carries the inverse unit of the applied perturbation by definition [55].

Next, we use the general ISF expression in Equation 4.8 to describe the first and second harmonic injection locking sensitivity of the device with precession given by Equation 4.4. The normalized trajectory is given by

$$\frac{\dot{\vec{m}}_{eq}}{|\dot{\vec{m}}_{eq}|^2} = \left[\frac{1}{2} \omega_g^2 (m_x^2 + 4m_y^2 + m_z^2) + \frac{1}{2} \omega_g^2 (m_x^2 - m_z^2) \cos(2\psi) - 2m_y^2 \omega_g^2 \cos(4\psi) \right]^{-1} \times \begin{pmatrix} m_x \omega_g \cos(\psi) \\ -2m_y \omega_g \sin(2\psi) \\ -m_z \omega_g \sin(\psi) \end{pmatrix} \quad (4.9)$$

The ISF for first harmonic injection locking by an rf current is

$$\Gamma_{1f}(t) = \frac{\vec{\tau}_{1f}}{V_{1f}(t)} \cdot \frac{\dot{\vec{m}}_{eq}}{|\dot{\vec{m}}_{eq}|^2} \quad (4.10)$$

where $\vec{\tau}_{1f}$ is given in Equation 4.2. Based on symmetry, the components of the spin polarization $\hat{p}$ contribute to different injection locking processes. The component perpendicular to the static equilibrium magnetization direction $\hat{B}_{ext}$, p_x , applies an antisymmetric driving force with respect to the oscillation orbit and therefore contributes to first harmonic locking [71]. The first harmonic torque term is then

$$\begin{aligned} \vec{\tau}_{1f} &= -\frac{\gamma_e \sigma}{R_{HM}} (\vec{m} \times (\vec{m} \times p_x \hat{x})) \\ &= \frac{\gamma_e \sigma}{R_{HM}} \sin(\gamma_p) \begin{pmatrix} (m_{y0} + m_y \cos(2\psi))^2 + m_z^2 \cos^2(\psi) \\ -m_x \sin(\psi) (m_{y0} + m_y \cos(2\psi)) \\ -m_x m_z \sin(\psi) \cos(\psi) \end{pmatrix} \end{aligned} \quad (4.11)$$

When Equation 4.10 is evaluated, the result reduces to a Fourier series

$$\Gamma_{1f}(t) = \frac{\omega_g \gamma_e \sigma \sin(\gamma_p)}{R_{HM} |\dot{\vec{m}}_{eq}|^2} \sum_{n=1,3,5} a_n \cos(n(\omega_g t + \phi)) \quad (4.12)$$

with non-zero coefficients

$$\begin{aligned} a_1 &= m_x \left(\frac{1}{2} m_y^2 + m_y m_{y0} + m_{y0}^2 + \frac{3}{4} m_z^2 \right) \\ &\quad + m_x m_{y0} m_y + \frac{1}{4} m_x m_z^2 \\ a_3 &= \frac{3}{4} m_x m_y^2 \\ a_5 &= -\frac{1}{4} m_x m_y^2 \end{aligned}$$

Equation 4.12 takes a similar form to the Fourier series of a square wave, as we see clearly when it is plotted in Figure 4-1. Intuitively, this is because the moment trajectory is nonzero along $\hat{x}$ for most of the cycle, so the first harmonic phase locking sensitivity remains large across the cycle.

The ISF for second harmonic injection locking follows from a similar analysis. Considering that the oscillator speed is effectively doubled, the ISF for second harmonic locking is

$$\Gamma_{2f}(t) = \frac{\vec{\tau}_{2f}}{V_{2f}(t)} \cdot \frac{\dot{\vec{m}}_{eq}}{4|\dot{\vec{m}}_{eq}|^2} \quad (4.13)$$

where $\vec{\tau}_{2f}$ is given in Equation 4.2. The spin polarization component parallel to the static equilibrium magnetization direction $\hat{B}_{ext}$, p_y , applies a symmetric driving force with respect to the equilibrium orbit and therefore contributes to second harmonic locking. The corresponding second harmonic torque term is

$$\begin{aligned} \vec{\tau}_{2f} &= -\frac{\gamma_e \sigma}{R_{HM}} (\vec{m} \times (\vec{m} \times p_y \hat{y})) \\ &= -\frac{\gamma_e \sigma}{R_{HM}} \cos(\gamma_p) \begin{pmatrix} -m_x \sin(\psi) (m_{y0} + m_y \cos(2\psi)) \\ m_x^2 \sin^2(\psi) + m_z^2 \cos^2(\psi) \\ -m_z \cos(\psi) (m_{y0} + m_y \cos(2\psi)) \end{pmatrix} \end{aligned} \quad (4.14)$$

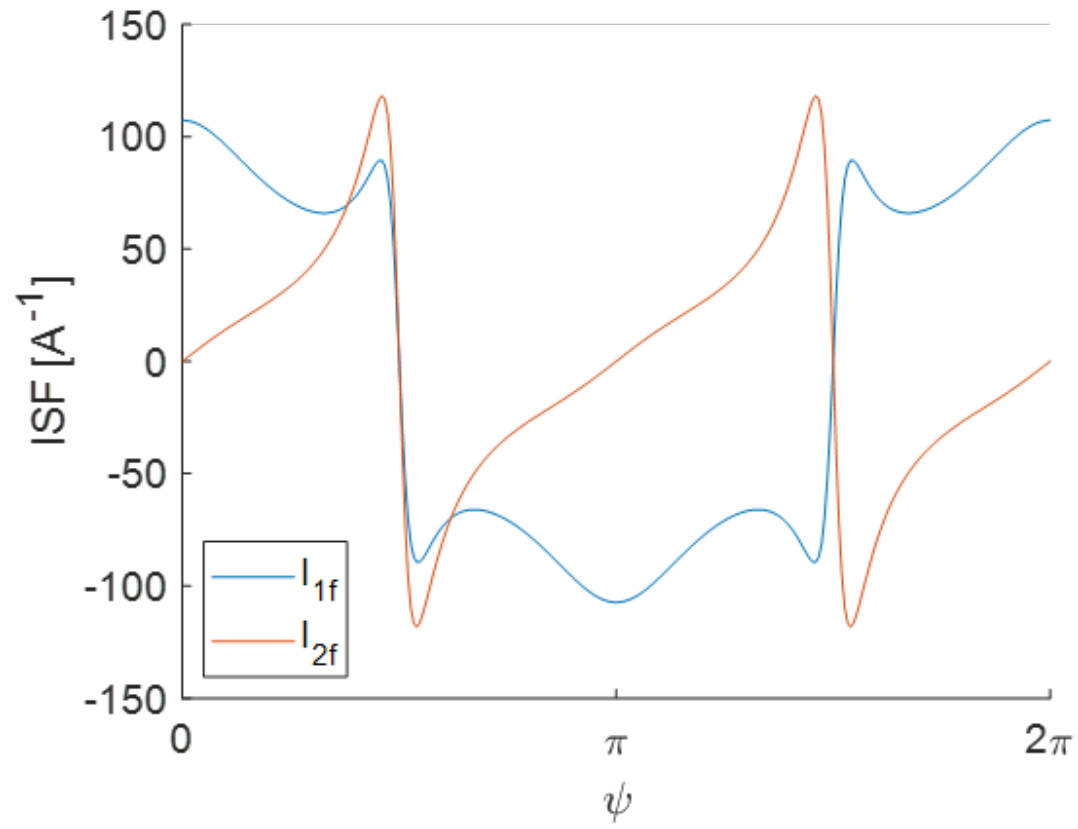


Figure 4-1: Impulse sensitivity functions for injection locking by first and second harmonic currents based on device with precession in Figure 3-4.

Upon evaluating Equation 4.13, the ISF reduces again to a Fourier series

$$\Gamma_{2f}(t) = \frac{\omega_g \gamma_e \sigma \cos(\gamma_p)}{4R_{HM}|\dot{\vec{m}}_{eq}|^2} \sum_{n=1,2} b_n \sin(2n(\omega_g t + \phi)) \quad (4.15)$$

with non-zero coefficients

$$b_1 = \frac{1}{2}m_x^2 m_{y0} + m_y(m_x^2 + m_z^2) - \frac{1}{2}m_z^2 m_{y0}$$

$$b_2 = \frac{1}{4}m_y(m_z^2 - m_x^2)$$

which looks similar to a sawtooth wave at the second harmonic. The second harmonic ISF is plotted in Figure 4-1. Because the moment trajectory is mainly along $\hat{y}$ near the "corners" of the orbit, where the oscillation speed $|\dot{\vec{m}}_{eq}|$ is lowest, there are large peaks in the ISF at $\psi = \frac{\pi}{2}, \frac{3\pi}{2}$ but the second harmonic phase locking sensitivity remains low throughout the rest of the cycle.

4.3 Correlation of Impulse Sensitivity Function with Adler's Equation

The ISF derived in the previous section can be used to construct the well-known Adler's equation to predict the injection locking characteristic of the spin torque oscillator [61]. While the ISF provides detailed insight into the phase sensitivity of the oscillator over each cycle, Adler's equation contains information about the macroscopic locking properties between the oscillator and a known external signal that we outline in this section.

Adler's equation can be found by taking the cross-correlation of the ISF and perturbing signal over the oscillator period T [75, 58]

$$\frac{d\phi(t)}{dt} = \omega_g - \omega_e + \frac{\omega_g}{T} \int_0^T \Gamma_\psi(\omega_g t + \phi) P_\psi(\omega_e t + \phi_e) dt \quad (4.16)$$

where we denote a change of argument from t to ψ for the ISF Γ and perturbing signal

P by the subscript ψ . Under combined first and second harmonic injection locking, the perturbing signal is given by

$$\begin{aligned} P(t) &= V_{1f}(t) + V_{2f}(t) \\ &= V_{1f} \sin(\omega_g t + \phi_{e1}) - V_{2f} \cos(2(\omega_g t + \phi_{e2})) \end{aligned} \quad (4.17)$$

In this case, the integral in Equation 4.16 will expand to have two terms representing first and second harmonic injection locking as we show in Equation 2.6. By averaging over the oscillator period in the cross-correlation, only components of the ISF and perturbing signal at the same integer-multiple of the oscillator frequency will yield a non-zero contribution to Adler's equation.

To solve Equation 4.16 exactly, the Fourier transforms of the first and second harmonic ISFs in Equations 4.12 and 4.15 must be calculated numerically due to the time-varying oscillator speed $|\dot{\vec{m}}_{eq}(t)|$ in the denominator. For analytical purposes, we represent the exact Fourier series representation of the first and second harmonic ISFs as

$$\begin{aligned} \Gamma_{1f}(t) &= \frac{\omega_g \gamma_e \sigma}{R_{HM}} \sin(\gamma_p) \sum_n a'_n \cos(n(\omega_g t + \phi)) \\ \Gamma_{2f}(t) &= \frac{\omega_g \gamma_e \sigma}{4R_{HM}} \cos(\gamma_p) \sum_n b'_n \sin(2n(\omega_g t + \phi)) \end{aligned}$$

Then, we can solve Equation 4.16 in a simple analytical form

$$\begin{aligned} \frac{d\phi(t)}{dt} &= \omega_g - \omega_e + \frac{\omega_g^3 \gamma_e \sigma V_{1f}}{2\pi R_{HM}} a'_1 \sin(\gamma_p) \int_0^{2\pi/\omega_g} \cos(\omega_g t + \phi) \sin(\omega_g t + \phi_{e1}) dt \\ &\quad - \frac{\omega_g^3 \gamma_e \sigma V_{2f}}{4\pi R_{HM}} b'_1 \cos(\gamma_p) \int_0^{\pi/\omega_g} \sin(2(\omega_g t + \phi)) \cos(2(\omega_g t + \phi_{e2})) dt \\ &= \omega_g - \omega_e - \frac{\omega_g^2 \gamma_e \sigma V_{1f}}{2R_{HM}} a'_1 \sin(\gamma_p) \sin(\phi - \phi_{e1}) - \frac{\omega_g^2 \gamma_e \sigma V_{2f}}{8R_{HM}} b'_1 \cos(\gamma_p) \sin(2(\phi - \phi_{e2})) \end{aligned} \quad (4.18)$$

Note that we neglect the time dependence of ϕ in the integral because the phase dynamics are assumed to occur on a much slower timescale than the oscillator period. Equation 4.18 is Adler's equation describing an oscillator injection locked by rf currents at the first and second harmonics with an explicit dependence of the respective injection locking strengths on the characteristics of the spin Hall oscillator.

If we extend to a coupled network of oscillators where the first harmonic signal is constructed from a superposition of other oscillator signals, Equation 4.18 is precisely Equation 2.6 with $A_{ij} = \omega_g^2 \gamma_e \sigma a'_1 V_j \sin(\gamma_p) / 2R_{HM}$ and $A_s = \omega_g^2 \gamma_e \sigma b'_1 V_{2f} \cos(\gamma_p) / 8R_{HM}$ where V_j is the rf signal amplitude from oscillator j and the second harmonic signal phase ϕ_{e2} is set to 0. We can write the corresponding interaction potential for the electrically coupled network of oscillators from Equation 2.7 using $-\frac{\partial H}{\partial \phi} = \frac{d\phi}{dt}$ [59]

$$H = - \sum_{i,j,i \neq j}^N \frac{\omega_g^2 \gamma_e \sigma V_j G_{ij}}{2} a'_1 \sin(\gamma_p) \cos(\phi_i - \phi_j) - \sum_i^N \frac{\omega_g^2 \gamma_e \sigma V_{2f}}{16R_{HM}} b'_1 \cos(\gamma_p) \cos(2\phi_i) \quad (4.19)$$

where V_j is the output signal of oscillator j and G_{ij} is the coupling conductance between oscillators i and j . This can be directly mapped to the Ising model with inter-node coupling coefficients $J_{ij} = \omega_g^2 \gamma_e \sigma V_j G_{ij} a'_1 \sin(\gamma_p) / 2$, which is indeed proportional to the inter-node conductance G_{ij} . Therefore, we have found an explicit relation between the characteristics of the spin Hall oscillator and the Ising Hamiltonian represented by an electrically coupled network of these oscillators.

4.3.1 Micromagnetic Validation of Locking Behavior

We validate the macroscopic injection locking behavior predicted by Adler's equation (Equation 4.16) against micromagnetic simulations of the spin Hall oscillator. The first and second harmonic components of Adler's equation for the spin Hall oscillator (Equation 4.18) with $V_{1f} = V_{2f} = 66$ mV are plotted separately in Figure 4-2.

From Adler's equation, we can predict a few important characteristics of the injection locking behavior for the spin Hall oscillator. First, the amplitude of Adler's equation $|\frac{d\phi}{dt}|$ gives the injection locking strength between the oscillator and external signal. The Ising machine solution time is expected to scale inverse linearly with the injection locking strength, so this can be a useful metric in comparing different oscillators for use in the Ising machine [14].

The injection locking strength is conventionally measured by the phase locking bandwidth $\Delta = |\omega_e - \omega_g| = |\frac{d\phi}{dt}|$ representing the maximum detuning frequency

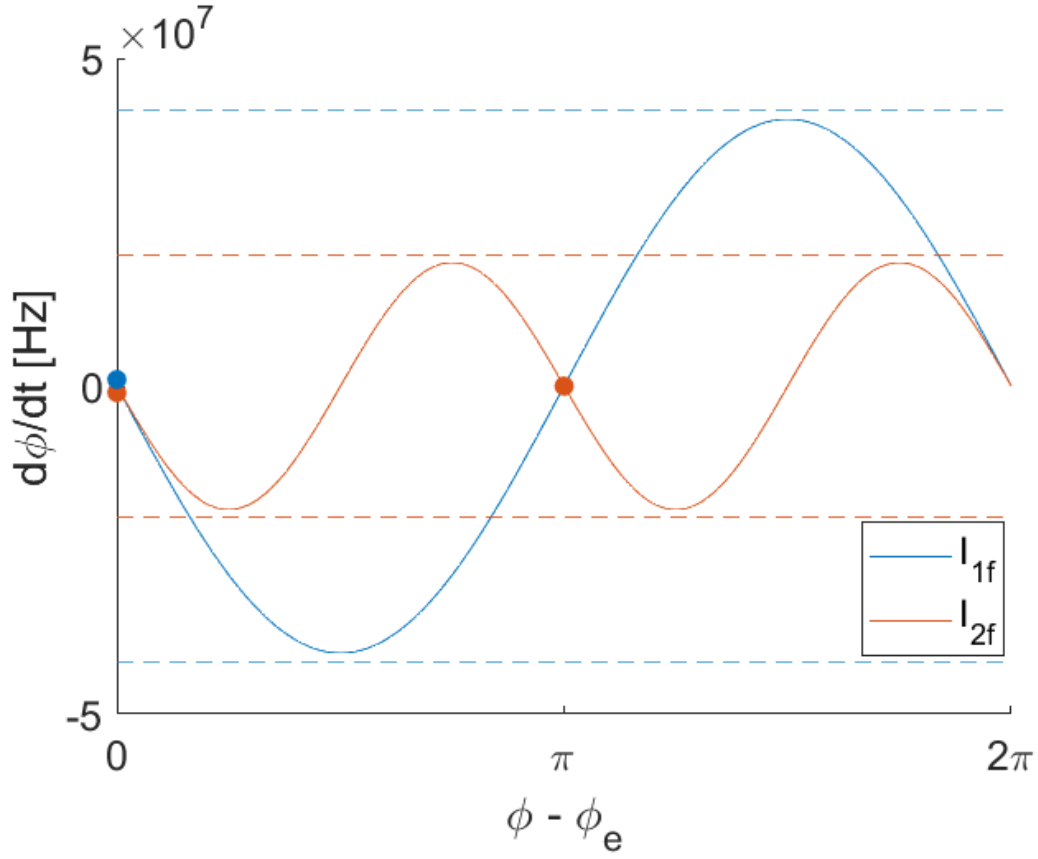


Figure 4-2: Numerically calculated Adler's equation for the spin Hall oscillator based on the ISFs in Figure 4-1 for $V_{1f}(t) = 66 \sin(\omega_g t)$ mV and $V_{2f} = -66 \cos(2\omega_g t)$ mV. The dotted lines mark the injection locking strengths calculated in micromagnetic simulations (Figures 4-3 and 4-4) that are expected to match the amplitudes $|d\phi/dt|$. The stable locking angles between the oscillator and external signals are marked by dots.

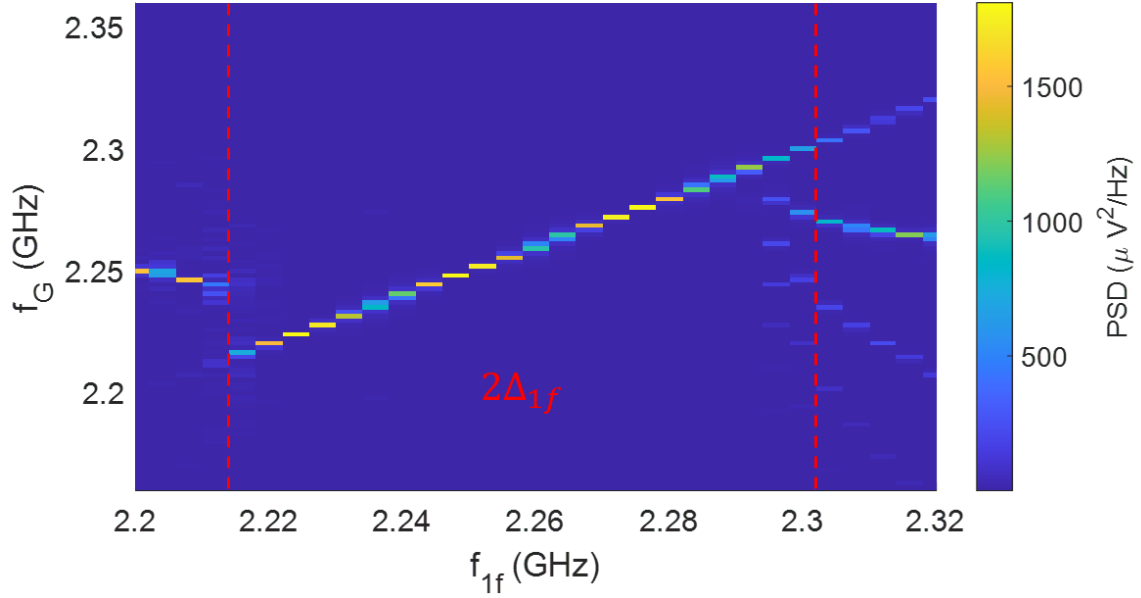


Figure 4-3: Power spectral density of the spin Hall oscillator under injection locking at the first harmonic ($f_{1f} \approx f_g$) calculated in micromagnetic simulations. Dotted lines mark the phase locking bandwidth $\Delta_{1f} = 2\pi \cdot 42$ MHz.

between the oscillator and external locking signal where phase locking can occur [61]. In micromagnetic simulations, we find the phase locking bandwidth of our spin Hall oscillator by applying an rf injection locking signal $V_{1f} = V_{2f} = 66 \sin(\omega_e t)$ mV with the frequency ω_e swept about ω_g and $2\omega_g$ for first and second harmonic locking, respectively. When the frequency detuning $|\omega_e - \omega_g|$ or $|\omega_e - 2\omega_g|$ is less than the first or second harmonic phase locking bandwidth Δ_{1f} or Δ_{2f} , the oscillator frequency is expected to follow ω_e , indicating a phase locked state. Outside the phase locking bandwidth, the oscillator phase is unlocked and the frequency returns to the free-running value ω_g .

The power spectral density of the oscillator's microwave output signal is plotted as a function of $f_e = \omega_e/(2\pi)$ in Figures 4-3 and 4-4 for first and second harmonic injection locking. The vertical red lines show the boundaries of the region where the oscillator frequency follows the external signal frequency, or the phase locking bandwidth. The phase locking bandwidths of the simulated device are compared to the corresponding Adler's equations in Figure 4-2 (marked by horizontal dotted lines)

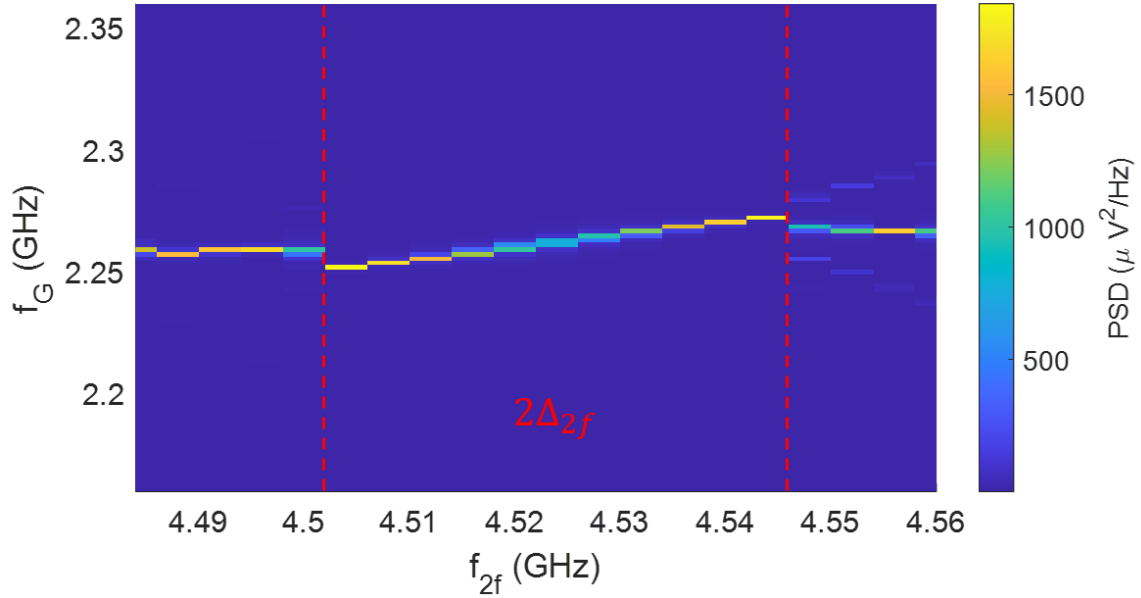


Figure 4-4: Power spectral density of the spin Hall oscillator under injection locking at the second harmonic ($f_{2f} \approx 2f_g$) calculated in micromagnetic simulations. Dotted lines mark the phase locking bandwidth $\Delta_{2f} = 2\pi \cdot 20$ MHz.

at the first and second harmonics. The numerically calculated amplitude of Adler's equation very closely matches the simulated phase locking bandwidth, showing that our ISF-based model can accurately predict the injection locking behavior at the first and second harmonics.

From Adler's equation, we can also tell the stable phase locking angles between the oscillator and external signal at points where $\frac{d\phi}{dt} = 0$ and $\frac{d^2\phi}{dt} < 0$ (marked by dots in Figure 4-2). At the first harmonic, the oscillator is expected to stably lock in-phase with the external signal, while second harmonic locking yields bistable locking angles at $[0, \pi]$. In a coupled network of oscillators subjected to a weak global second harmonic signal, this bistable locking will lead to binarization of the oscillator phases as described in Section 2.3.

4.4 Impulse Sensitivity Function for Phase Noise

The derivation of the ISF for magnetic field perturbation follows from a similar analytical procedure as described in Section 4.2 for the spin torque ISF. For magnetic

fields directed along $\hat{x}$, $\hat{y}$, or $\hat{z}$, the ISF is given by the general form

$$\Gamma_{Bi}(t) = \frac{\vec{\tau}_{Bi}}{B_i} \cdot \frac{\dot{\vec{m}}_{eq}}{|\dot{\vec{m}}_{eq}|^2} \quad (4.20)$$

where $i = (x, y, z)$. The relevant torque terms from Equation 4.2 are the first and second terms, but the first term is much larger in the case of low magnetic damping α so the second term is neglected. The torque due to a perturbing magnetic field (considered separate from $\vec{B}_{eff}$) is then

$$\vec{\tau}_{Bi} = -\gamma_e(\vec{m} \times \vec{B}_i) \quad (4.21)$$

Explicitly in x, y , and z , these torque terms are

$$\begin{aligned} \vec{\tau}_{Bx} &= -\gamma_e \begin{pmatrix} 0 \\ m_z \cos(\psi) \\ -m_{y0} - m_y \cos(2\psi) \end{pmatrix} \\ \vec{\tau}_{By} &= -\gamma_e \begin{pmatrix} -m_z \cos(\psi) \\ 0 \\ m_x \sin(\psi) \end{pmatrix} \\ \vec{\tau}_{Bz} &= -\gamma_e \begin{pmatrix} m_{y0} + m_y \cos(2\psi) \\ -m_x \sin(\psi) \\ 0 \end{pmatrix} \end{aligned}$$

As in the case of rf current, the ISFs can be represented by Fourier series

$$\Gamma_{Bx}(t) = \frac{\omega_g \gamma_e}{2|\dot{\vec{m}}_{eq}|^2} \sum_{n=1,3} b_n'' \sin(n\psi) \quad (4.22)$$

$$\Gamma_{By}(t) = \frac{\omega_g \gamma_e}{|\dot{\vec{m}}_{eq}|^2} m_x m_z \quad (4.23)$$

$$\Gamma_{Bz}(t) = \frac{\omega_g \gamma_e}{2|\dot{\vec{m}}_{eq}|^2} \sum_{n=1,3} a_n'' \cos(n\psi) \quad (4.24)$$

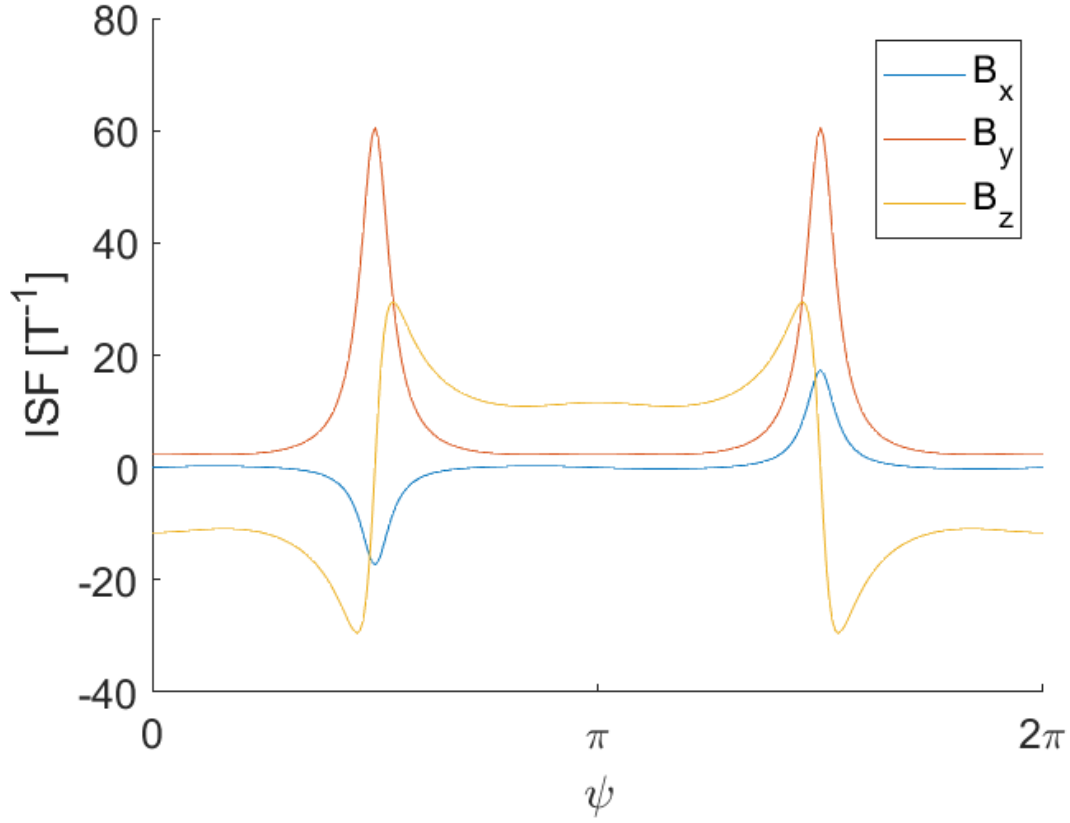


Figure 4-5: Numerically calculated impulse sensitivity function for magnetic field perturbations along x, y, z axes.

with non-zero Fourier coefficients

$$b_1'' = (3m_y - 2m_{y0})m_z$$

$$b_3'' = m_y m_z$$

$$a_1'' = -(3m_y + 2m_{y0})m_x$$

$$a_3'' = m_x m_y$$

The ISFs in Equations 4.22-4.24 are plotted in Figure 4-5. We see that the highest sensitivity to external field perturbations occurs at $\psi = \pi/2, 3\pi/2$, or at the slow “corners” of the oscillation orbit. Elsewhere in the cycle, the sensitivity is relatively low.

In the macrospin approximation, thermal fluctuations are commonly represented

by an effective magnetic field (Section 5.1.4). Therefore, the ISF for perturbations by an external magnetic field can also be used to represent the oscillator's time-varying sensitivity to thermally induced phase noise [26, 18]. In the following chapter, we use this ISF method to implement phase noise in our oscillator device macromodel. While it is not a focus of this thesis, we also note that one may use the ISF as a tool to analytically predict and optimize the phase noise performance of the oscillator device [18].

Chapter 5

Circuit-Level Modeling

We develop a lightweight, versatile device model in Verilog-A that can efficiently emulate the spin Hall oscillator’s nonlinear phase dynamics across a range of operating points. In this chapter, we describe the development of the device model and its integration into circuit-level simulations of the Ising machine.

5.1 Verilog-A Oscillator Device Model

Because of recent interest in using spin torque oscillators for computing applications, circuit-level spin torque oscillator macromodels have been proposed to emulate the oscillator characteristics with varying levels of detail. One such approach involves implementing equivalent circuits in SPICE-based circuit simulators to solve the macrospin LLGS equation (Equation 3.1) for the spin torque oscillator [17, 54, 6]. While this approach can model the oscillator’s characteristics with a reasonable degree of accuracy by simulating the fundamental equation governing the magnetic dynamics, the computational workload is heavy and leads to long simulation times. For our goal of simulating large-scale coupled arrays of spin torque oscillators at the circuit level, this approach is infeasible.

A second approach involves the implementation of high-level analytical equations governing the spin torque oscillator behavior in Verilog-A [3, 12, 80, 34, 5]. Because these models use a higher level of abstraction than the LLGS equation, the simulation

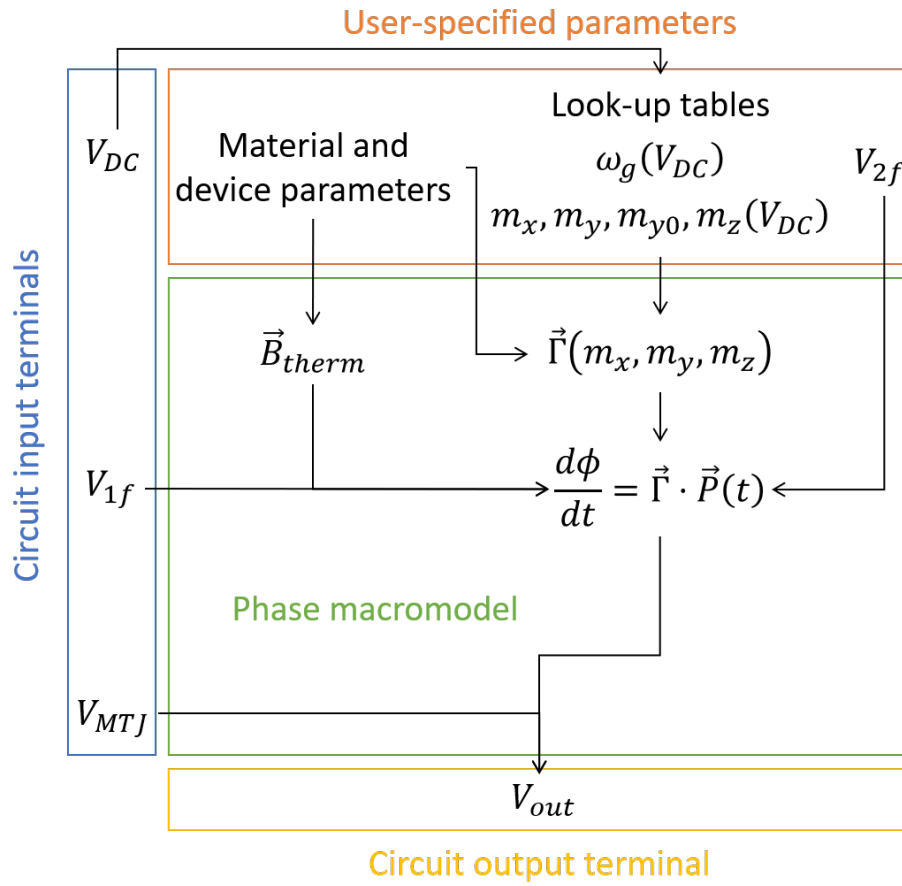


Figure 5-1: Schematic of the oscillator device macromodel implemented in Verilog-A.

time is more reasonable than the first approach. However, so far, comprehensive analytical equations exist only for describing the general electrical characteristics of the spin Hall oscillator such as the nonlinear dependence of frequency, microwave output power, and linewidth on the driving force [61]. Complete analytical models of injection locking for the spin Hall oscillator have not yet been analytically derived, and as such, there are no existing device macromodels that can simulate this behavior in a detailed and computationally efficient way.

Based on the analytical model developed in Chapter 4, we choose to implement a lightweight ISF-based phase macromodel that enables efficient simulation of the oscillator injection locking behavior [56, 25, 43, 78, 49]. Our analytical ISF formulas are integrated into the device macromodel to allow for continuous modeling of the injection locking behavior across a range of DC operating points.

A schematic of the spin Hall oscillator macromodel implemented in Verilog-A is shown in Figure 5-1. In the following subsections, we describe each component of the device model in further detail.

5.1.1 Input-Output Terminals

The 3-terminal spin Hall oscillator is modeled in Verilog-A as a 3-port device. Two input ports are reserved for applying the DC excitation V_{DC} and first harmonic coupling signal $V_{1f}(t)$ to the heavy metal layer. Compared to the real physics of the spin Hall oscillator where DC and rf torques acting on the precessing moment can be treated independently, the phase model we use cannot distinguish between the two, so they must be applied at separate ports. The DC voltage is also used to set the operating point of the device that determines the free-running frequency and oscillation orbit. Finally, the global second harmonic voltage $V_{2f}(t)$ is integrated into the device model as a user-specified parameter.

The third device port specifies the input and output signals at the MTJ terminal. A DC voltage V_{MTJ} can be applied to the MTJ nanopillar to read out the microwave

signal V_{out} by TMR, modeled by the equation

$$V_{out} = \frac{V_{MTJ}}{R_{MTJ}} \left(R_{MTJ} + \frac{R_{AP} - R_P}{2} m_x \sin(\omega_g t + \phi(t)) \right) \quad (5.1)$$

where $R_{MTJ} = (R_{AP} + R_P + R_{HM})/2$ is the equivalent DC resistance between the MTJ readout terminal and ground terminal (ground is implicitly included in the Verilog-A device). The variable definitions and numerical values used in our device are described in Table 5.1 and in the following section.

5.1.2 Device Parameters

To generalize our macromodel to a wide range of spin torque oscillator devices and operating points, we allow the user to specify material and device parameters that determine the oscillator’s behavior. User-specified constants, as well as the values we use, are summarized in Table 5.1.

The device can be modeled at various DC operating points by specifying the oscillation frequency ω_g and components of the equilibrium precession $\vec{m}_{eq}$ (to first order, as in Equation 4.4) as look-up tables indexed by V_{DC} . These values can be extracted from a one-time micromagnetic or macrospin simulation of the spin Hall oscillator and supplied to the Verilog-A device model. To further abstract the physics from the device model, we can consider directly interfacing the Verilog-A code with macrospin or micromagnetic simulations so that the user is not required to manually extract the relevant parameters.

5.1.3 Nonlinear Phase Macromodel

Our nonlinear phase macromodel numerically solves Adler’s equation for the oscillator phase based on the ISF and incoming perturbing signal at each simulation timestep [18, 56, 25, 43, 78, 49]. Because Adler’s equation is just a first order differential equation, it can be solved very efficiently in SPICE-based circuit simulators. The

exact form that is implemented in the device model is given below

$$\frac{d\phi(t)}{dt} = \omega_g \vec{\Gamma}(\omega_g t + \phi(t)) \cdot \vec{P}(\omega_e t + \phi_e(t)) \quad (5.2)$$

where $\vec{\Gamma}(t)$ is the ISF and $\vec{P}(t)$ is the perturbing signal. Inside the Verilog-A device, Equation 5.2 can be equivalently represented by a simple capacitor with the $i - v$ relation $\frac{dv_c}{dt} = \frac{1}{C} i_c$ where the capacitor voltage v_c represents the phase $\phi(t)$ and the right-hand side is calculated by the multiplication of the free-running frequency, ISF, and perturbation at each discrete simulation timestep. This modeling approach is remarkably compact, as the complete nonlinear injection locking dynamics of the oscillator can be represented by the simple equation of a capacitor. The model's simplicity is made possible by the fact that the ISF contains all necessary information about the spin torque oscillator's physics to describe injection locking. Although the detuning factor $\omega_g - \omega_e$ in Adler's equation (Equation 2.6) is not explicitly written in Equation 5.2 because the external signal frequency ω_e is not strictly known, detuning effects are implicitly included in the interaction of the ISF and perturbation. Therefore, this simple method completely models the macroscopic injection locking behavior of the spin Hall oscillator.

The ISF and perturbation in Equation 5.2 are represented by vectors where one component is reserved for each independent perturbation mechanism acting on the oscillator: our model includes the components $\{V_{1f}, V_{2f}, B_x, B_y, B_z\}$. To analytically calculate the ISF in Verilog-A, the DC voltage V_{DC} is sampled in the initial simulation step to extract ω_g and the components of $\vec{m}_{eq}$ from the given look-up tables. These parameters are used as arguments in the ISF formulas (Equations 4.12, 4.15, and 4.22-4.24) that are implemented in the device macromodel. The ISF is calculated analytically only once over the course of the simulation, just after the DC operating point has been established. While in this thesis we have only derived the analytical equations for the ISF in the case of the clamshell-shaped precession, this modeling approach can be generalized to other precession orbits by deriving the relevant form of the ISF.

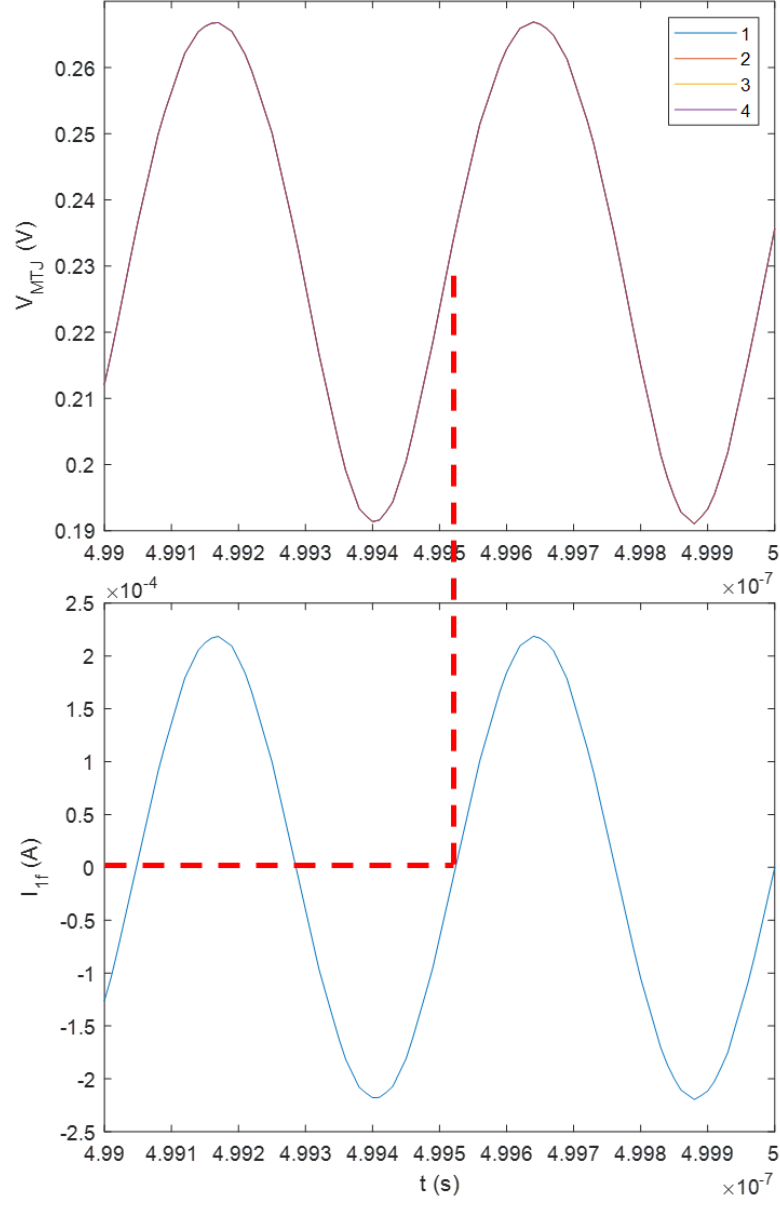


Figure 5-2: Injection locking of the Verilog-A spin torque oscillator at the first harmonic. As predicted by Adler's equation, the oscillator signal V_{out} synchronizes in-phase with the applied first harmonic signal V_{1f} .

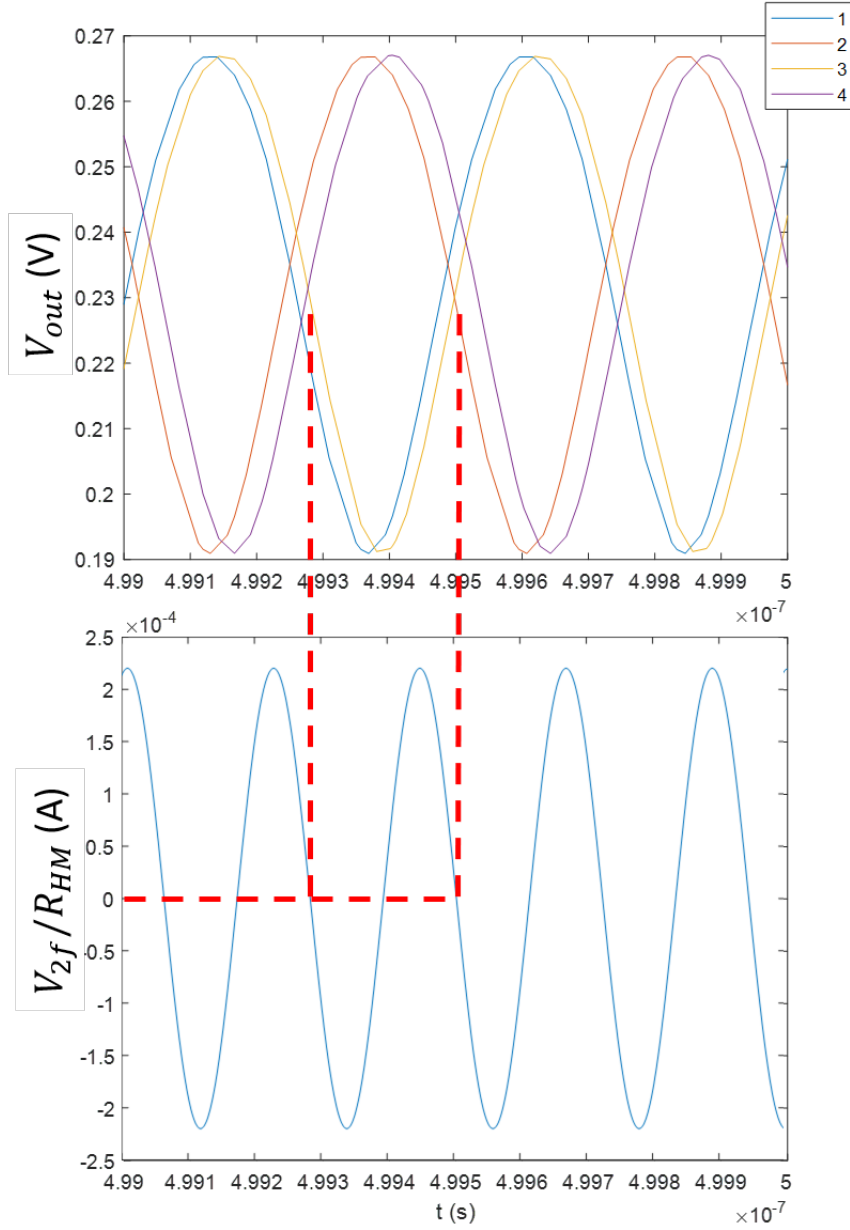


Figure 5-3: Injection locking of the Verilog-A spin torque oscillator at the second harmonic. In this case, the oscillator signal V_{out} synchronizes bistably with the applied second harmonic signal V_{2f} . Recall that the in-phase locking was predicted to occur with the minus cosine wave at the second harmonic (Figure 4-2), as is demonstrated here.

As a simple demonstration of the ISF-based phase macromodel's behavior, we reproduce the stable locking angles predicted by Adler's equation (Figure 4-2) by modeling injection locking in Verilog-A. We run 4 independent trials of locking the oscillator to first and second harmonic signals with a random initial phase. At the first harmonic, we confirm the monostable locking behavior where the oscillator signal settles in-phase with the sinusoidal signal (Figure 5-2). At the second harmonic, we observe bistable locking at $\phi = [0, \pi]$ (Figure 5-3) where, at each stable locking point, the oscillator signal locks at the same point on the second harmonic signal as predicted by Adler's equation (Figure 4-2).

5.1.4 Thermal Phase Noise

As we described briefly in Section 4.4, the ISF for magnetic field perturbations can be used to describe phase noise resulting from stochastic thermal fluctuations [49]. In this section, we describe how thermal phase noise is implemented in our Verilog-A phase macromodel.

Thermal fluctuations can be treated as an effective thermal field in the macrospin approximation [9, 77]

$$\vec{B}_{therm} = \vec{\eta}(t) \sqrt{\frac{2\alpha k_B T}{\gamma_e V M_s \Delta t}} \quad (5.3)$$

where $\vec{\eta}(t)$ is a randomly generated 3D vector with random normally distributed x , y , and z coordinates, T is the temperature, and V is the oscillator free layer volume. Every Δt units of time, new components of $\vec{\eta}(t)$ are randomly generated. Based on this random vector and specified material parameters in Table 5.1, the x , y , and z components of the thermal field are calculated in the Verilog-A device model and multiplied by the corresponding ISFs Γ_{Bx} , Γ_{By} , and Γ_{Bz} to calculate the resulting phase noise as described by Equation 5.2.

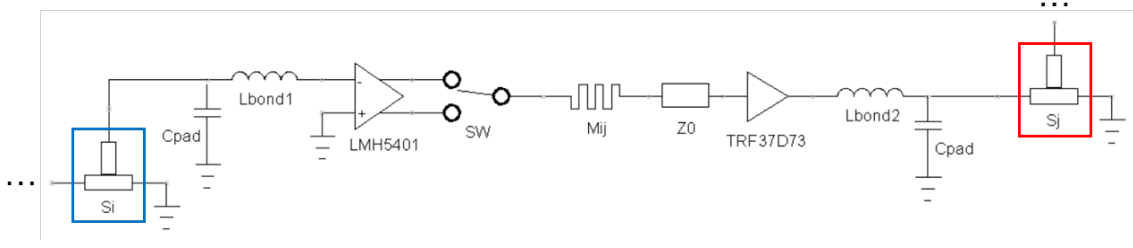


Figure 5-4: Schematic of HSPICE coupling circuit from oscillator i (blue) to j (red). Wirebond parasitics are modeled by $C_{pad} = 100$ fF and L_{bond} assuming an inductance/length of 1 nH/mm ($L_{bond1} = 4$ nH, $L_{bond2} = 8$ nH). Auxiliary gain-setting resistors and DC biasing elements are omitted for visual clarity. Fan-out/in to/from other coupled oscillators occur at the LMH outputs and TRF input, respectively.

5.2 Electrically Coupled Spin Hall Oscillator Array

Next, we integrate our Verilog-A oscillator macromodel into HSPICE circuit simulations to demonstrate the phase dynamics of a small-scale electrically coupled oscillator network. To show that our phase macromodel accurately emulates the coupled spin Hall oscillator dynamics, we also model a small network in micromagnetic simulations and compare the temporal phase dynamics. We show that in a small 4-node MAX-CUT example problem simulated in HSPICE and micromagnetic simulations, the final phase configuration of the coupled oscillator network settles to the ground state of the Ising Hamiltonian under a binarizing second harmonic signal.

5.2.1 HSPICE Circuit Model

One branch of the electrical coupling circuit implemented in HSPICE is shown in Figure 5-4. The output of the oscillator is buffered by Texas Instruments LMH5401, and a secondary rf gain block (Texas Instruments TRF37D73) ensures an adequate coupling signal amplitude. The fully differential buffer amplifier provides oscillator signals of both positive and negative polarities to the coupling network, allowing problems with coupling coefficients of both signs to be modeled. The coupling conductance is set by a series voltage-programmable memristor and transmission line. The transmission line delay is set to compensate the net phase shift around the coupling loop caused by wirebond and amplifier parasitics. The total phase shift between the MTJ output and

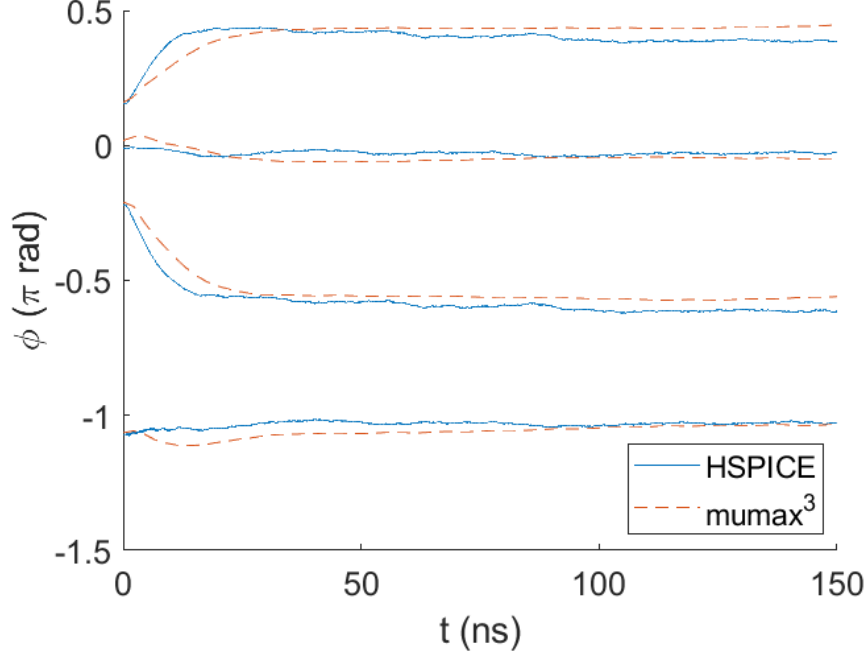


Figure 5-5: 4-coupled oscillator network solving Ising model with uniform inter-oscillator coupling $J_{ij} = -1$. Phase dynamics are simulated in mumax³ and HSPICE with first harmonic coupling only ($V_{MTJ} = 0.2\text{V}$).

heavy metal input terminals should be zero to avoid modulations of the frequency and coupling strength based on finite delay time [68, 70, 67, 35].

An all-to-all coupled network of 4 oscillators (shown abstractly in Figure 3-2) is simulated in HSPICE to establish a baseline result. Explicitly, the coupling signal applied to each oscillator in the coupled network is given by

$$V_{cpl,i} = A_{amp} \sum_{j, i \neq j}^4 \frac{R_{HM}}{R_{Mij} + Z_0} \frac{V_{MTJ}}{R_{MTJ}} \frac{R_{AP} - R_P}{2} m_x \sin(\omega_g t + \phi_j) \quad (5.4)$$

where $A_{amp} = 5.5 \text{ V/V}$ is the total signal gain around the coupling loop, $1/(R_{Mij} + Z_0)$ is the coupling conductance, and only the rf component of $V_{out,j}$ is included because the coupling circuit filters DC signals.

We solve a benchmark MAX-CUT problem with uniform unit weighting $J_{ij, i \neq j} = -1$ ($R_{Mij} + Z_0 = 150\Omega$) where the ground state phase configuration of the 4 oscillators corresponds to any permutation of $\{\phi_1, \phi_2, \phi_3, \phi_4\} = \{0, 0, \pi, \pi\}$. Without second

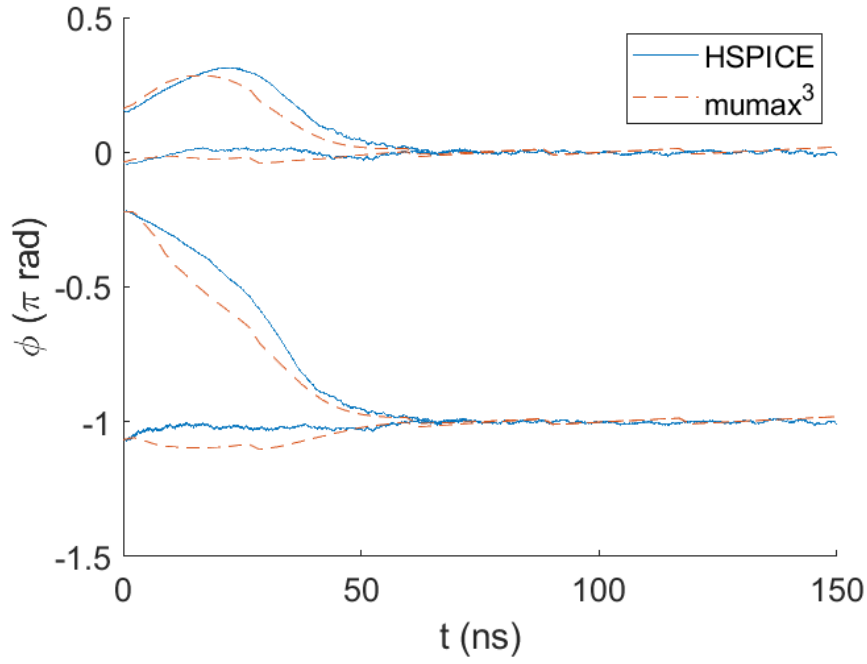


Figure 5-6: 4-coupled oscillator network solving Ising model with uniform inter-oscillator coupling $J_{ij} = -1$. Phase dynamics are simulated in mumax³ and HSPICE with first harmonic coupling ($V_{MTJ} = 0.2\text{V}$) and second harmonic locking ($V_{2f} = -4.0 \cos(2\omega_g t)\text{V}$).

harmonic locking (Figure 5-5, solid curves), the phases settle almost uniformly across 0 to 2π , and the Ising Hamiltonian ground state is not clearly visible. However, once a second harmonic signal is applied (Figure 5-6, solid curves), the phases clearly binarize and settle to the ground state configuration.

5.2.2 Coupled Oscillators in Micromagnetic Simulation

To demonstrate that our circuit model accurately emulates the coupled oscillator dynamics, we model an equivalent system in micromagnetic simulations and compare the phase dynamics. Modifications are made to the open-source mumax³ code to enable electrical signal generation and coupling between multiple oscillators.

Micromagnetic Source Code Modification

The open-source mumax³ code¹ is modified to enable electrical signal generation and coupling of up to 4 oscillators. The 4 oscillators are simulated within a single strip geometry with dimensions $6\mu\text{m} \times l_{MTJ} \times t_{FL} \text{ nm}^3$ and are each spaced apart by $2\mu\text{m}$ to minimize coupling via the stray fields. The discretization cell size remains $5 \times 5 \times 1.5 \text{ nm}^3$ in x, y, and z.

The new quantities defined in the micromagnetic simulation source code are shown in Table 5.2. Both numerical parameters and magnetization-dependent quantities are introduced to model the electrical characteristics and coupling of the spin Hall oscillators. Numerical parameters are defined uniformly across the 4 modeled oscillators. For the definitions of the magnetization-dependent quantities, the 4 magnetic oscillators are assigned to separate material regions with indices 1-4. In mumax³, each region is modeled with different material and excitation parameters, which allows for the independent calculation of the magnetoresistance and coupling signal at each oscillator.

First, to construct the electrical output signal of each oscillator, the oscillator magnetoresistance is introduced as a region-wise magnetization-dependent quantity

$$R_{out} = \frac{R_P - R_{AP}}{2} m_x + \frac{R_P + R_{AP}}{2} \quad (5.5)$$

¹<https://github.com/mumax/3>

based on the parallel R_P and anti-parallel R_{AP} magnetoresistance parameters and assuming a fixed layer polarized along $\hat{x}$. R_{out} is updated at each simulation timestep based on the region's m_x value from the previous timestep using customized versions of the mumax³ numerical solvers.

To construct the coupling signal $J_{cpl,i}$ delivered to oscillator i , a weighted sum of the other oscillator signals $j \neq i$ is calculated based on the coupling conductances contained in the vector $\overline{G}$. Explicitly, the coupling signal to oscillator i is

$$J_{cpl,i} = J_{read} \sum_{j=1, j \neq i}^4 G_{ij} \frac{R_P - R_{AP}}{2} m_{x,j} \quad (5.6)$$

where the inter-oscillator coupling is assumed to be reciprocal $G_{ij} = G_{ji}$. Because circuit gain elements are not modeled, J_{read} implicitly contains any necessary gain factor to reach an appropriate coupling signal magnitude. In addition, the DC part of R_{out} is dropped because the coupling circuit is assumed to filter DC signals.

The coupling current density $J_{cpl,i}$ is applied to oscillator i by adding to the existing region-dependent spin transfer torque term in mumax³. In contrast to the regular implementation of STT, this current term depends on magnetization and is reevaluated and updated at each timestep through R_{out} to capture the dynamically changing phase of each oscillator.

Electrical Coupling Benchmark in Micromagnetic Simulation

Using micromagnetic simulations, we model the same MAX-CUT problem as we did in HSPICE (Section 5.2.1). The same device parameters and excitations are applied as listed in Table 5.1 (V_{DC} and V_{MTJ} are mapped to equivalent current densities by $J = V/(R_{HM} \cdot w_{HM} \cdot t_{HM})$). In Figures 5-5 and 5-6, the phase dynamics simulated in micromagnetic simulations (dashed lines) are plotted with the previously shown HSPICE results (solid lines). In Figure 5-7, we show spatial plots of the in-plane oscillator magnetization with and without second harmonic locking corresponding to the points $t = 150$ ns in Figures 5-5 and 5-6.

The key features of the phase dynamics, including time-to-solution and final steady-

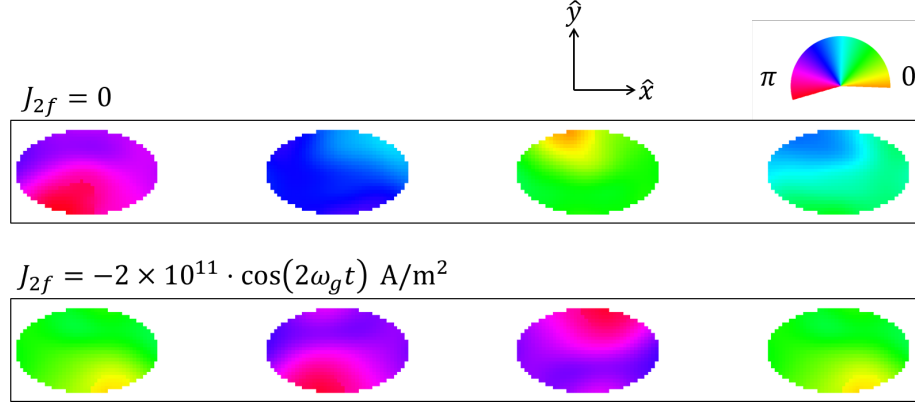


Figure 5-7: Spatial plots of the in-plane free layer magnetization simulated in mumax without second harmonic locking (top) and with second harmonic (bottom). The color key maps the in-plane magnetization to the oscillator phase. The plotted data is taken at $t = 150$ ns from the mumax³ simulation results in Figures 5-5 and 5-6. The inter-oscillator spacing has been reduced for visibility.

state phase configuration, are very well matched between micromagnetic and circuit simulations. Slight discrepancies in the shapes of the phase curves can be attributed to the significantly simplified ISF-based modeling used in HSPICE compared to the physically detailed mumax³ dynamics. Nevertheless, the circuit model accurately reproduces the macroscopic phase dynamics of the coupled oscillator network from micromagnetic simulations and can be simulated in a small fraction of the time.

Table 5.1: User-specified device parameters in the Verilog-A spin Hall oscillator macromodel at the specified operating point V_{DC} , V_{MTJ} . The values given are those used in our simulations.

Parameter	Symbol	Value	Unit
Parallel MTJ resistance	R_P	1200	Ω
Anti-parallel MTJ resistance	R_{AP}	2400	Ω
Heavy metal resistance	R_{HM}	200	Ω
Heavy metal width	w_{HM}	200	nm
Heavy metal thickness	t_{HM}	5	nm
Spin polarization angle	γ_p	30	$^\circ$
Spin Hall angle	θ_{SH}	0.07	
Gilbert damping	α	0.02	
Saturation magnetization	M_s	800	kA/m
MTJ width	w_{MTJ}	120	nm
MTJ length	l_{MTJ}	200	nm
Free layer thickness	t_{FL}	1.5	nm
DC voltage	V_{DC}	88	mV
MTJ voltage	V_{MTJ}	200	mV
Free-running frequency	ω_g	$2\pi \cdot 2.26$	rad GHz
$\vec{m}_{eq}$ components	m_x m_y m_{y0} m_z	0.99 0.34 0.6 0.21	
Temperature	T	0 or 300	K
Thermal fluctuation timestep	Δt	15	ps

Table 5.2: Numerical parameters and magnetization-dependent quantities defined in the mumax³ source code for modeling electrical coupling of spin Hall oscillators.

Numerical Parameter	Symbol	Value	Unit
Parallel MTJ resistance	R_P	1200	Ω
Anti-parallel MTJ resistance	R_{AP}	2400	Ω
MTJ read current density	J_{read}	6.7×10^8	A/m ²
Coupling conductance vector	$\overline{G}$	$\{G_{12}, G_{13}, G_{14}, G_{23}, G_{24}, G_{34}\}$	-
$\vec{m}$ -Dependent Quantity	Symbol	Type	Unit
Oscillator magnetoresistance	R_{out}	Scalar Value	Ω
Coupling current density	J_{cpl}	Excitation	A/m ²

Chapter 6

Statistical Performance of the Ising Machine

In this section, we use the HSPICE circuit model of the Ising machine to analyze its statistical performance on problems with various coupling coefficients, array sizes, and phase noise. We solve MAX-CUT problems on undirected ($J_{ij} = J_{ji}$) weighted and unweighted graphs with a circular architecture, shown in Figure 5-4.

6.1 Coupling Coefficient Bitweight

First, we study the performance of the Ising machine on solving size-16 weighted MAX-CUT problems with mixed-sign graph edge weights $w_{ij} = -J_{ij}$. We vary the bit resolution of the coupling coefficients from 1-6 bits (in the 6-bit case, the memristor values vary over $R_{Mij} \approx 50\Omega - 3.4k\Omega$). Problems represented by coefficients with higher bit resolutions are closer to being analog and therefore have a more anisotropic energy landscape, making them more difficult to solve. A constant second harmonic signal $V_{2f} = -30 \cos(2\omega_g t)$ mV is used on all the problems, and thermal phase noise is neglected at first ($T = 0$ K).

In Figure 6-2, we show the probability of reaching the Ising Hamiltonian ground state (GS) and within the 3 higher states above GS ($\leq \text{GS}+3$) as a function of the coupling coefficient bit resolution. The ground state of the Ising Hamiltonian represents

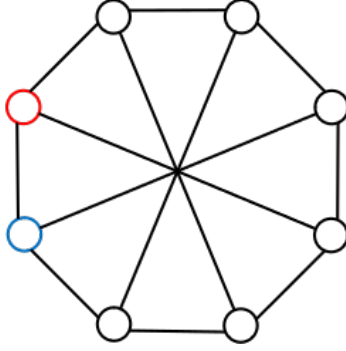


Figure 6-1: Schematic of the circular graph architecture used in system-level simulation of the Ising machine for $n = 8$ nodes. Graph nodes represent oscillators (red and blue nodes correspond to red and blue oscillators in Figure 5-4) and edges represent coupling between oscillators using conductances determined by R_{Mij} and Z_0 .

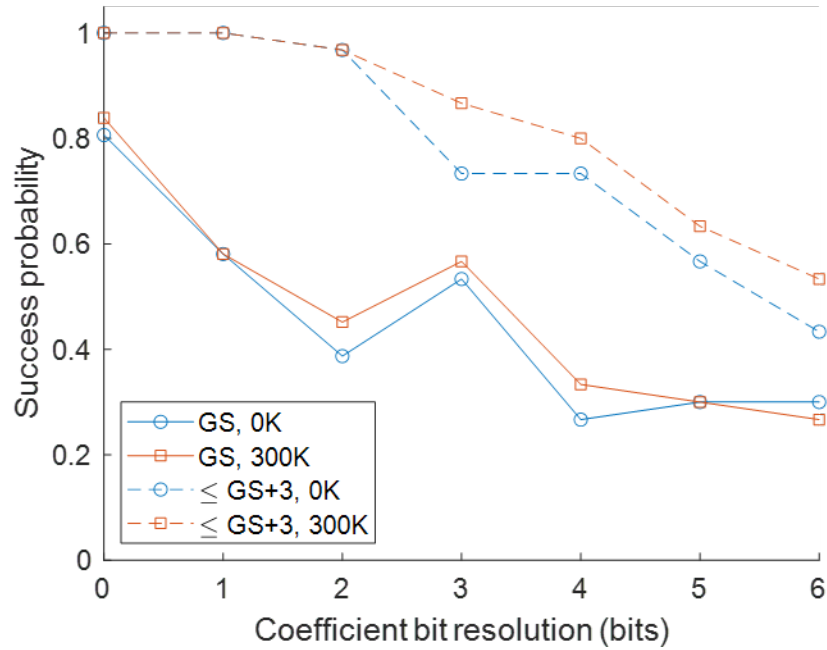


Figure 6-2: Probability of reaching the ground state and next-highest-energy state vs. coupling coefficient bit resolution at $T = 0K$ (solid lines) and $T = 300K$ (dashed lines).

the optimal solution of the MAX-CUT problem, while increasingly higher states above GS represent increasingly sub-optimal solutions. The data is averaged over more than 500 independent runs with randomly generated coupling coefficients and initial oscillator phase configurations. As expected due to the increasingly analog coefficients, the GS and GS+3 success probabilities decrease for higher bit resolutions as it becomes more difficult to find a close-to-optimal solution.

6.2 Phase Noise Impact on Performance

We perform the same simulations as in Section 6.1 considering thermal phase noise at $T = 300$ K. The results are shown by orange lines in Figure 6-2. For coefficient bit resolutions above 2 bits, both the GS and GS+3 probabilities increase by up to 13% under the influence of phase noise. Below 2 bits, the presence of phase noise makes a negligible difference in both probabilities. Also, at one higher bit resolution (coefficient bitweight=6), the GS probability is slightly lower with phase noise than without. One possible reason for this could be that the selected second harmonic signal amplitude is too large for problems with higher-resolution bitweights. Further studies will be needed to provide insight on the optimal second harmonic signal amplitude or annealing schedule for different problems [20].

Figure 6-3 provides more insight into the trend of increasing success probability in the presence of phase noise. We plot a histogram of the final energy states over all simulations in Figure 6-2 at $T = 0$ K and $T = 300$ K. Without phase noise ($T = 0$ K), the final energy states are concentrated in the two lowest energy states (0=GS and the next highest state 1), but the probability distribution spreads out across the higher-energy states. On the other hand, when thermal phase noise is included, the state probabilities are most highly concentrated within the first few energy states and the probability monotonically decreases with higher energy states. In addition, the highest sub-optimal state that the phases settle to is GS+22, whereas without phase noise, the phases get stuck in states as high as GS+33. This suggests that thermal phase noise helps the oscillator phases to escape local energy minima and more closely

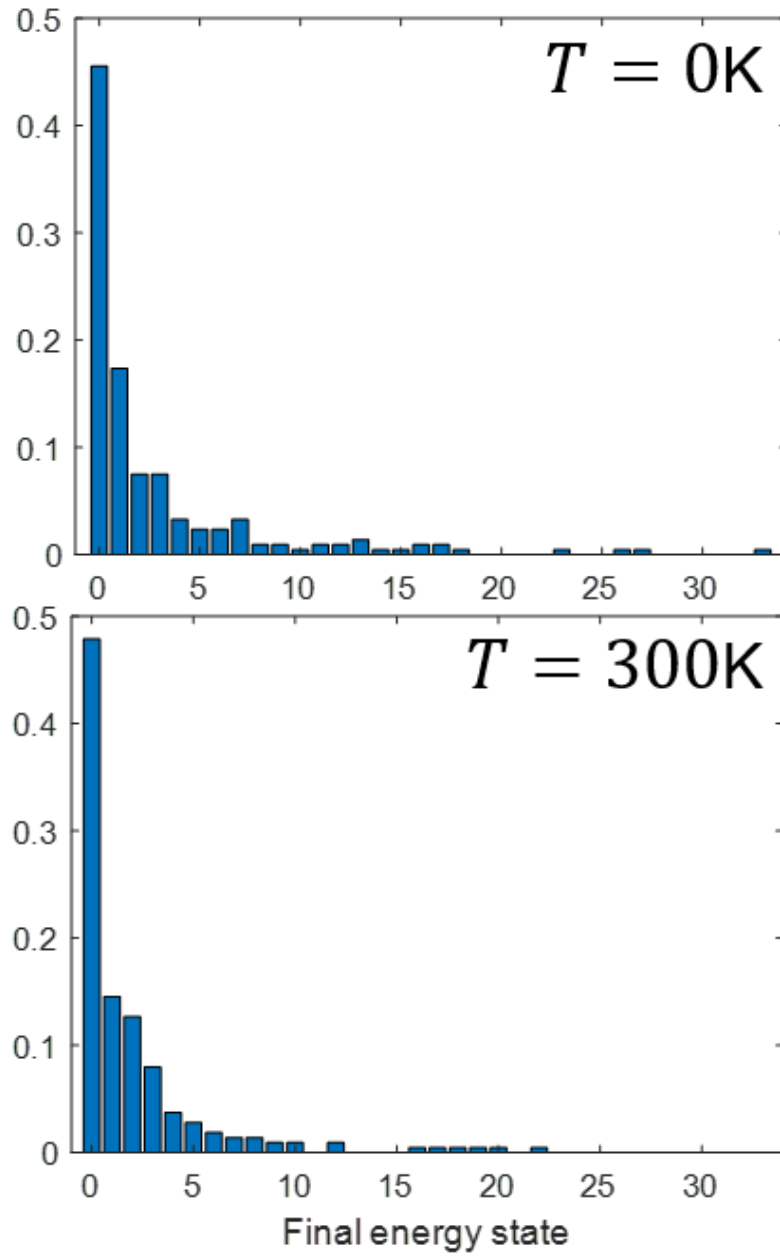


Figure 6-3: Histograms of final energy states for 16-node MAX-CUT problems with randomly generated coupling coefficients at $T = 0\text{ K}$ and $T = 300\text{ K}$.

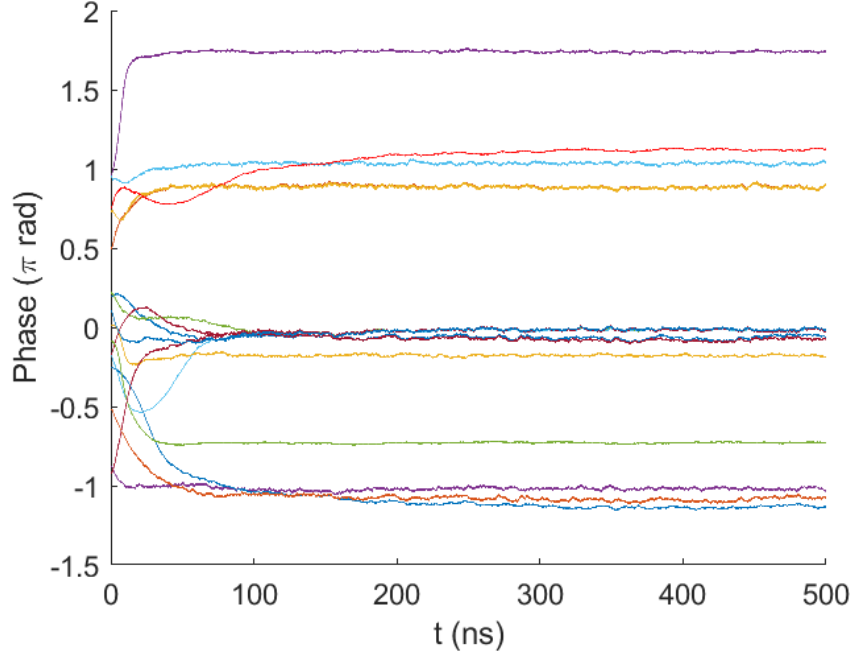


Figure 6-4: Oscillator phases vs. time solving a 16-node problem with randomly generated coefficients in the absence of phase noise.

approach the ground state solution of the Ising Hamiltonian.

In Figures 6-4 and 6-5, we plot the temporal phase dynamics of the Ising machine solving a 16-node MAX-CUT problem in the absence and presence of phase noise. In Figure 6-4 at $T = 0$ K, the phases stabilize by $t = 50$ ns but are stuck in a local energy minimum above the global minimum GS. In Figure 6-5 at $T = 300$ K, the phases are similarly initially stuck in a local energy minimum above GS. However, due to destabilization by phase noise, there is a phase slip event around $t = 150$ ns where two of the oscillators change their phase by π to bring the system to a lower-energy state. This demonstrates how phase noise can help the coupled array to reach lower-energy states by escaping local energy minima.

Because we have found thermal phase noise to be beneficial to the Ising machine performance so far, it may be worth considering alternative oscillator operating points. Closer to the auto-oscillation threshold, the oscillator nonlinearity is stronger and leads to enhanced injection locking strength [60]. This enhanced injection locking strength could lead to faster solution times for the oscillator-based Ising machine. However,

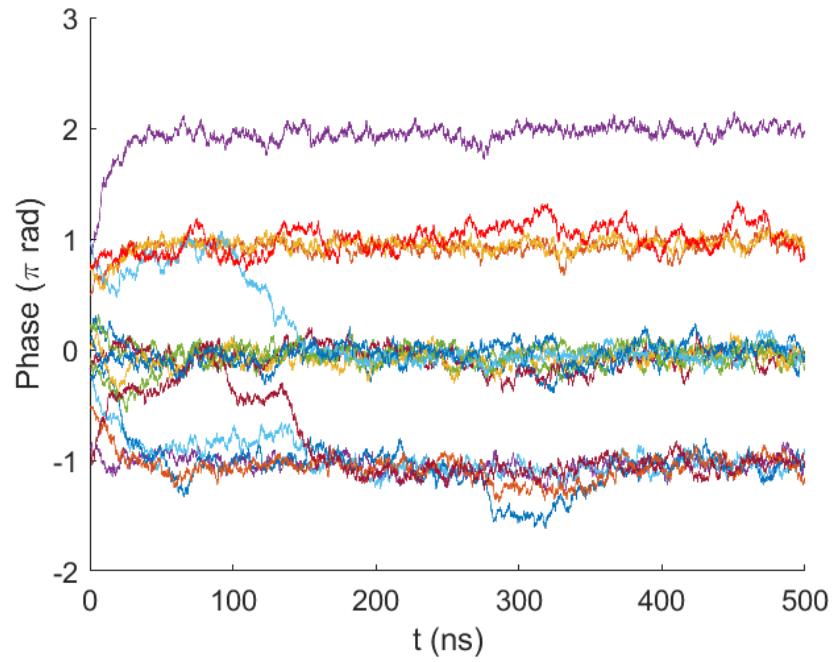


Figure 6-5: Oscillator phases vs. time solving a 16-node problem with randomly generated coefficients in the presence of phase noise. The same problem is simulated in Figure 6-4, but in the presence of phase noise, the phases reach a lower-energy configuration.

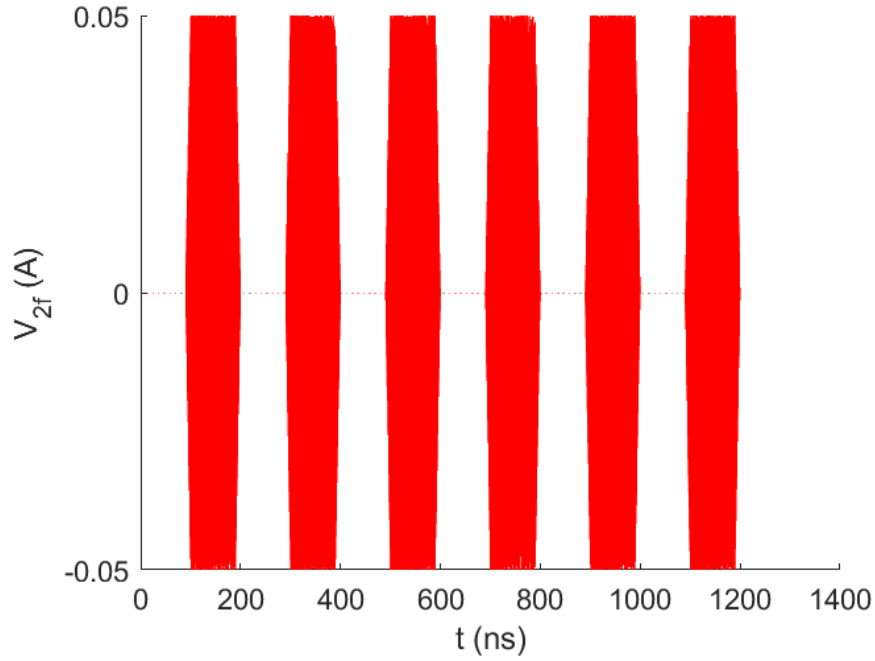


Figure 6-6: Annealing schedule employed in scaling simulations of the Ising machine.

the increased nonlinearity introduces more complexities in the coupling dynamics, such as increased thermal phase noise due to stronger coupling between the oscillator amplitude and phase [36]. It is unknown whether these semi-chaotic dynamics would improve or degrade the performance of a computing system using coupled array of spin torque oscillators such as the Ising machine.

6.3 Scaling Trends with Coupled Array Size

Next, we demonstrate the scaling trends of the ground state probability and solution time of the spin Hall oscillator Ising machine with the number of coupled oscillators n . We use coupled arrays of sizes $n = 16 - 256$ on unweighted graphs with uniform coupling coefficients $J_{ij} = -1$. Because the phase binarization process is slower and more difficult for larger arrays, an second harmonic annealing schedule is used where the amplitude of V_{2f} is stepped between 0 and 50 mV every 100 ns, resulting in a strong binarization of the phases [75]. The annealing schedule $V_{2f}(t)$ is shown in Figure 6-6 and the response of the oscillator phases with $n = 256$ to the annealing

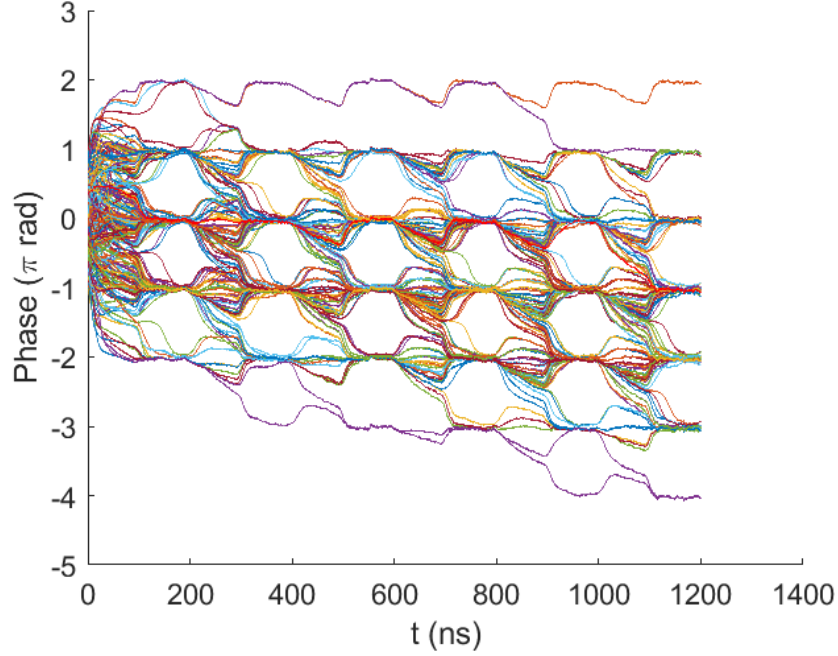


Figure 6-7: Response of the oscillator phases to the annealing schedule in Figure 6-6 with the number of oscillators $n = 256$.

schedule is shown in Figure 6-7.

In Figure 6-8, we show the probability of reaching GS and a close-to-optimal solution (within 5% of GS) as a function of n . A total of 200 problems are simulated with randomly generated initial oscillator phase states. Though the GS probability falls to only about 13% by $n = 256$, the phases settle in close-to-optimal configurations (within 5% of GS) in nearly 90% of runs even up to the largest simulated array $n = 256$. Figure 6-9 shows a histogram of the final energy state distribution for $n = 256$. The final phase states are distributed among a few local energy minima close to GS ($E=-380$, blue dashed line) and across all simulated trials, the phases always settle within 90% GS (yellow dashed line).

In Figure 6-10, we show the ground state solution time of the spin Hall oscillator Ising machine as a function of n . The ground state solution time $t_{sol} = t_{GS}/P_{GS}$ is defined as the solution time of runs that reached the ground state (t_{GS}) multiplied by the expected number of runs needed to find the ground state ($1/P_{GS}$ where P_{GS} is the ground state probability in Figure 6-8). The solution times of a simulated 1 MHz

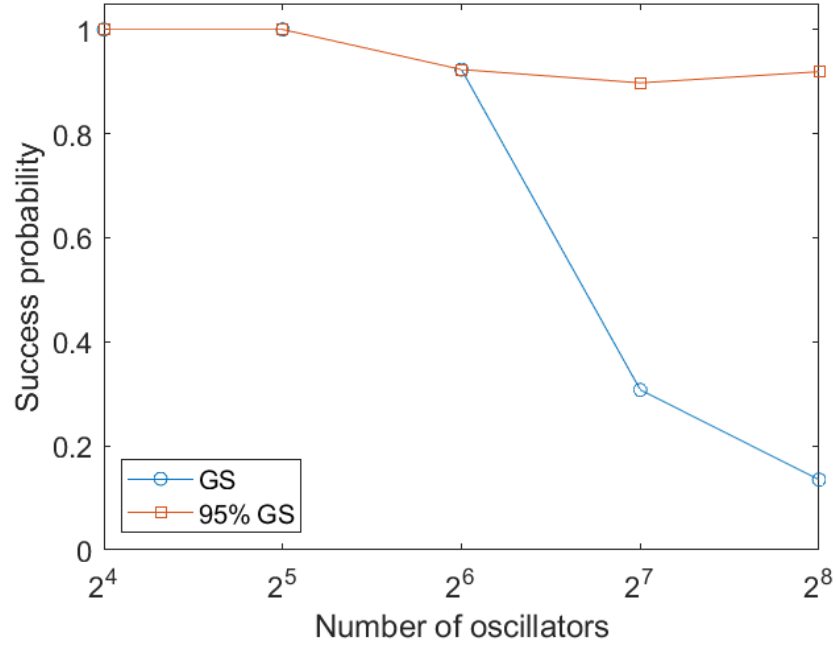


Figure 6-8: Probability of reaching ground state (GS) and within 95% of GS vs. number of oscillators n .

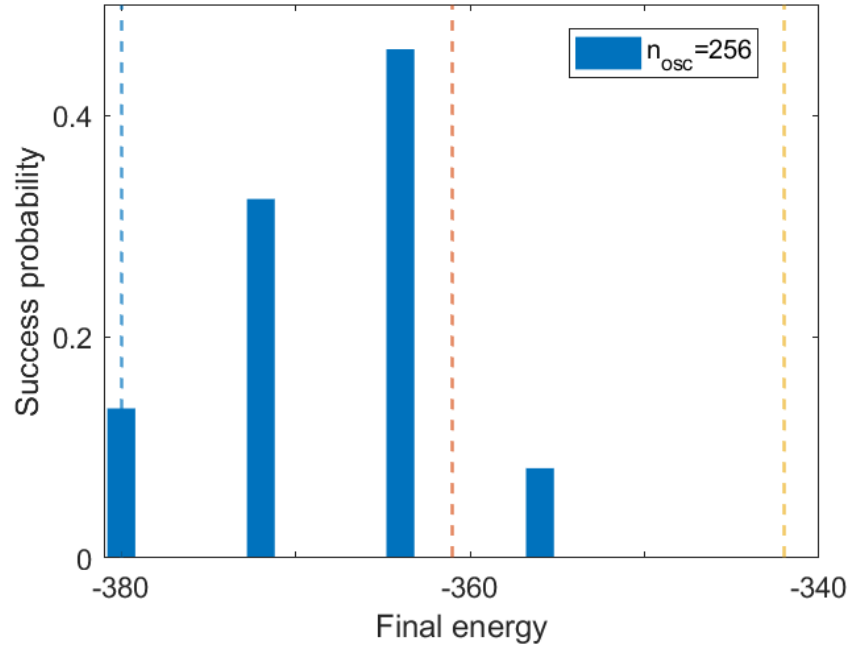


Figure 6-9: Histogram of final energy states for $n = 256$. From left to right, the vertical lines represent 1.0, 0.95, 0.9 GS.

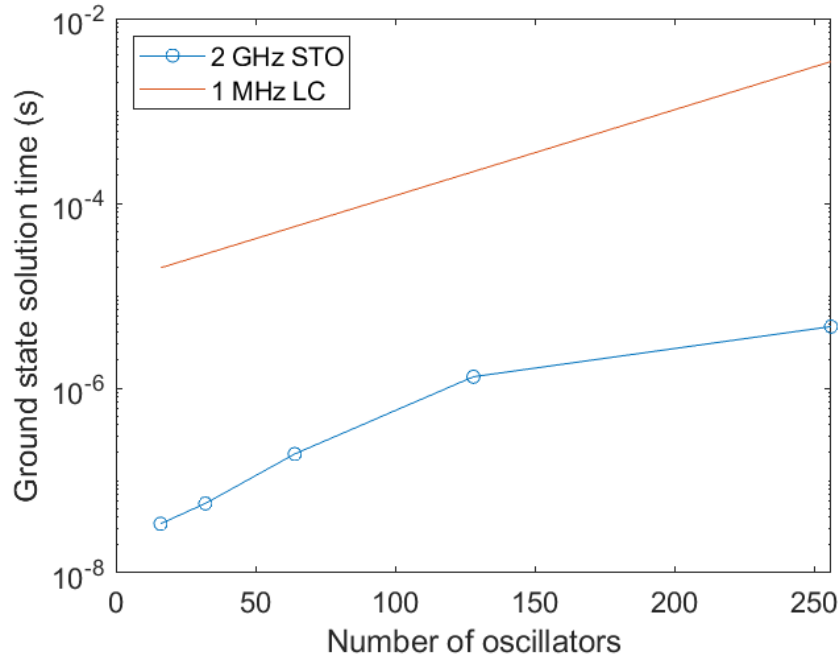


Figure 6-10: Time-to-solution t_{sol} vs. n . The orange line represents the expected solution time of a 1 MHz LC oscillator Ising machine reproduced from Chou *et al.*¹ [14].

¹Licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>)

LC oscillator Ising machine from [14] are also shown in Figure 6-10 for comparison. Both implementations follow an approximately logarithmic scaling trend of solution time with n . However, the GHz spin Hall oscillator Ising machine solution times are 3 orders lower than the MHz LC oscillator array due to the much higher oscillator operating frequency.

6.4 Energy Consumption

The fast solution time is arguably the most significant benefit of the spin Hall oscillator Ising machine, as it enables solving larger and more difficult problems on reasonable timescales and with less energy. In Table 6.1, we compare the solution time and energy consumption of our simulated GHz Ising machine against other simulated and experimentally realized Ising machines. We neglect the large power consumption due to off-the-shelf electronic components in our implementation. By using custom-designed amplifiers to interface with the spin Hall nano-oscillator, we can minimize the active power of the electronic components and relax the strict frequency limitation of about 2 GHz to achieve even higher performance [10].

Table 6.1: Performance comparison of the spin Hall oscillator Ising machine in this work to previous implementations of the Ising machine. Time and energy values are standardized to size-100 problems solved on cubic graphs.

	CPU [21]	GPU [37]	D-WAVE [8]	OPA [53]	CMOS [14, 13]	PTNO [21, 22]	SHNO
Solution time	246ms	100 μ s	30ms	2ms	2.5ms	25.5 μ s	0.5 μ s
Power	60W	$\leq$ 250W	25kW	-	2.5mW	1.2mW	96 μ W
Energy to solution	14.8J	$\leq$ 25mJ	750J	-	6.3 μ J	30.6nJ	48pJ
Energy efficiency (sol/s/W)	6.7×10^{-2}	$\geq$ 40	1.3×10^{-3}	-	1.6×10^5	3.26×10^7	2×10^{10}
Area (one element)	-	-	-	-	0.1mm ²	0.2 μ m ²	0.024 μ m ²
Room temp.	Y	Y	N	Y	Y	Y	Y
All-to-all connectivity	Y	Y	N	Y	Y	Y	Y

Chapter 7

Conclusion

7.1 Summary and Thesis Contributions

In this thesis, we have demonstrated the potential speed and energy benefits of the GHz spin Hall nano-oscillator Ising machine through analytical modeling and numerical simulations. First, we developed an analytical model based on a characteristic function of the oscillator called the impulse sensitivity function that can describe the detailed injection locking behavior of a spin Hall oscillator with any arbitrary precession orbit. Our analytical model is more general than existing descriptions of injection locking in spin torque oscillators and can predict the injection locking behavior at the oscillator's harmonics [61, 60]. Using this model, we developed a versatile and compact Verilog-A device model that can emulate the nonlinear injection locking dynamics of the spin Hall oscillator continuously across a range of DC operating points. We integrated this device model into circuit-level simulations that can be scaled up to large arrays of coupled oscillators to model the electrically coupled Ising machine.

By analyzing the statistical behavior of the Ising machine in circuit-level simulations, we demonstrated a few positive and interesting properties of the Ising machine using spin Hall oscillators. First, the ground state solution times roughly followed the expected logarithmic scaling trend with the size of the coupled array, but the solution times were about 3 orders of magnitude lower than those of a MHz LC oscillator Ising machine [14]. At the same time, the GHz-frequency Ising machine maintained a

high solution accuracy, arriving withing 5% of the ground state energy in almost 90% of trials even up to the largest simulated array size of 256 oscillators when solving MAX-CUT problems on unweighted graphs. Finally, we showed that the intrinsic thermal phase noise of the spin Hall oscillator is expected to help the phases reach to energy states closer to the ground state, further enhancing the Ising machine performance.

By studying the detailed behavior of the spin Hall oscillator-based Ising machine and offering insights into the desirable device topology and characteristics, this thesis offers helpful guidelines on the design of a GHz-frequency Ising machine. The analytical models and quantitative tools that we have introduced will enable the future experimental realization of a fast, energy-efficient, and scalable spin Hall oscillator-based Ising machine.

7.2 Future Work

The most obvious next step for this work is to pursue the experimental realization of the spin Hall oscillator-based Ising machine. However, experimental demonstrations of computing systems employing coupled arrays of spin torque and spin Hall oscillators are still limited [57, 69, 28]. Before a spin Hall oscillator-based Ising machine can be realized, advancements must be made in the synchronization of large arrays of magnetic oscillators.

Experimental work toward realizing robust electrical coupling between spin Hall nano-oscillators will be most useful to realize a general Ising machine architecture, as this mechanism will allow for all-to-all coupling between a large array of oscillators. All-to-all coupling is most favorable to model a wide range of combinatorial optimization problems including those with complex mappings to the Ising Hamiltonian [48]. If a voltage-controlled oscillator (VCO) based calibration scheme can be realized in an array of three-terminal spin Hall oscillators by taking advantage of their high degree of tunability, this could potentially mitigate the frequency spread that commonly limits the large-array synchronization of magnetic oscillators. Even if a small-scale Ising

machine is realized using an electrically coupled network of spin Hall oscillators, this would represent a significant step toward realizing a GHz-frequency Ising machine.

Further exploration of the Ising machine behavior with different annealing schedules and devices operating in different dynamical regimes will also be helpful. Because spin torque and spin Hall oscillators are a new technology proposed for the Ising machine based on significantly different physics than the previously employed electronic LC oscillator, there may be an optimal design of the annealing schedule that can achieve higher ground state solution probabilities. In addition, oscillators operating in highly nonlinear dynamical regimes have interesting characteristics including strong injection locking and enhanced phase noise [61, 40]. While we have shown in this work that a weakly nonlinear spin Hall nano-oscillator can achieve fast solution times and good solution probabilities for the Ising machine, the implications of using devices with more complex nonlinear dynamics in oscillator-based computing systems remains unknown.

Appendix A

Oscillator Design Considerations

We use our knowledge of the ISF to propose the optimal spin Hall oscillator device design for the Ising machine. We focus on optimizing the shape of the equilibrium orbit to maximize the first and second harmonic injection locking efficiencies. We sweep the DC voltage applied to the heavy metal to tune the precession from small-to large-angle and vary the perpendicular magnetic anisotropy field B_{PMA} to tune the out-of-plane precession component.

In Figure A-1, we show the generation frequency (top row), normalized magnetization components (middle row), and phase locking bandwidth normalized to frequency (bottom row) for two devices: one with PMA that reduces the out-of-plane demagnetizing field by about half (left column) and one without PMA (right column). The lower limit of the DC voltage range is chosen at the small-angle oscillation, where m_y remains almost along $\hat{B}_{ext}$, and the upper limit is chosen where the m_x oscillation is at the maximum ± 1 for the large-angle regime.

First, the phase locking effect is stronger in the small-angle oscillation regime. This is likely due to the fact that the spin polarization $\hat{p}$ has a large component along $-\hat{y}$, so the cross product in $\vec{\tau}_{1f}$ will be maximized when the moment remains almost along the applied field direction. In the near-threshold regime (small-angle oscillation), the phase locking effect is also expected to be enhanced by stronger nonlinearity that is not included in our model [61]. However, this enhanced nonlinearity will also introduce nonideal effects such as strong amplitude-phase coupling [36]. To avoid

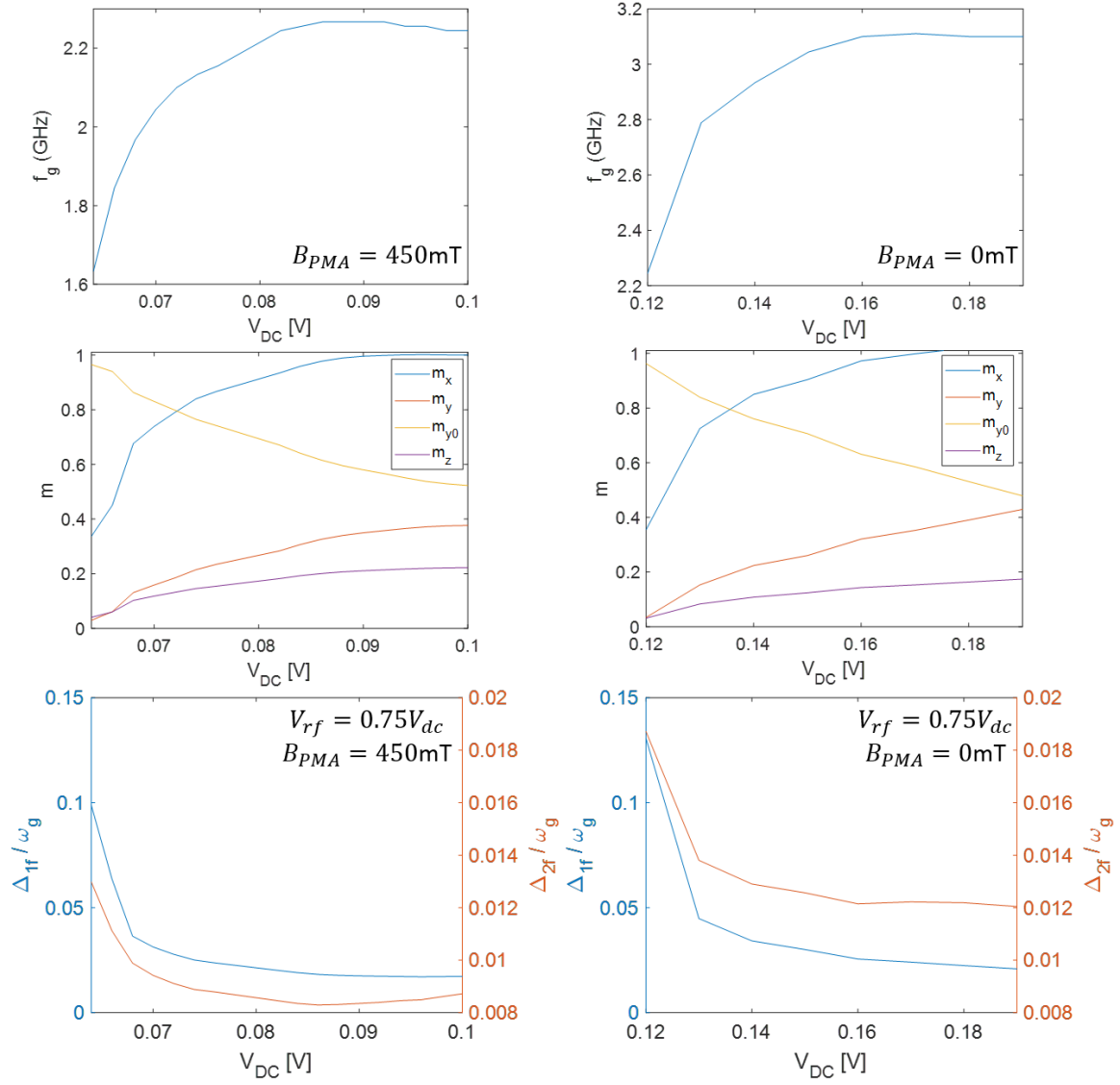


Figure A-1: Study of injection locking efficiency based on V_{DC} and the perpendicular magnetic anisotropy field B_{PMA} .

these nonidealities, we still favor an operating point $V_{DC}/V_{th} \approx 1.5$ where V_{th} is the threshold voltage for observing magnetization auto-oscillations.

Second, the first and second harmonic injection locking strengths are both enhanced in the oscillator without PMA. This is because the most significant components of $\vec{\tau}_{1f}$ and $\vec{\tau}_{2f}$ are in the in-plane $\hat{x}$ and $\hat{y}$ directions, respectively (Equations 4.11 and 4.14). The torque will couple to the oscillator phase when the trajectory is also along $\hat{x}$ or $\hat{y}$, so a more in-plane precession in a device with a larger out-of-plane demagnetizing field (less PMA) will enhance the injection locking effect. The second harmonic injection locking is especially enhanced by in-plane oscillation because the "corners" of the orbit are where the most m_z variation occurs, as well as where the second harmonic ISF is maximum. Therefore, second harmonic injection locking is more significantly affected by the decreased m_z component of the oscillation orbit.

In summary, a spin Hall oscillator without PMA is the most favorable device to maximize electrical coupling efficiency based on the largely in-plane oscillation orbit. As an added benefit, because of the higher effective field B_{eff} due to the larger out-of-plane demagnetizing field, the oscillation frequency will be increased in a device without PMA, contributing to even greater injection locking strength. While in subsequent parts we use the device with moderate PMA ($B_{PMA} = 450\text{mT}$) due to the frequency limit of our electronic circuit components, we can expect performance improvements if a device with less PMA is used for the Ising machine.

Bibliography

- [1] Phys. Rev. Lett. 83, 1834 (1999) - Spin Hall Effect.
- [2] R. Adler. A Study of Locking Phenomena in Oscillators. *Proceedings of the IRE*, 34(6):351–357, June 1946. Conference Name: Proceedings of the IRE.
- [3] Sora Ahn, Hyein Lim, Hyungsoon Shin, and Seungjun Lee. Analytic Model of Spin-Torque Oscillators (STO) for Circuit-Level Simulation. *JSTS:Journal of Semiconductor Technology and Science*, 13(1):28–33, February 2013.
- [4] Dagur Ingi Albertsson, Mohammad Zahedinejad, Afshin Houshang, Roman Khymyn, Johan Åkerman, and Ana Rusu. Ultrafast Ising Machines using spin torque nano-oscillators. *Applied Physics Letters*, 118(11):112404, March 2021. Publisher: American Institute of Physics.
- [5] Dagur Ingi Albertsson, Mohammad Zahedinejad, Johan Åkerman, Saul Rodriguez, and Ana Rusu. Compact Macrospin-Based Model of Three-Terminal Spin-Hall Nano Oscillators. *IEEE Transactions on Magnetics*, 55(10):1–8, October 2019. Conference Name: IEEE Transactions on Magnetics.
- [6] Sebastian Ament, Nikhil Rangarajan, Arun Parthasarathy, and Shaloo Rakheja. Solving the stochastic Landau-Lifshitz-Gilbert-Slonczewski equation for monodomain nanomagnets : A survey and analysis of numerical techniques. *arXiv:1607.04596 [cs]*, January 2017. arXiv: 1607.04596.
- [7] P. Andreani and Xiaoyan Wang. On the phase-noise and phase-error performances of multiphase LC CMOS VCOs. *IEEE Journal of Solid-State Circuits*, 39(11):1883–1893, November 2004. Conference Name: IEEE Journal of Solid-State Circuits.
- [8] Sergio Boixo, Troels F. Rønnow, Sergei V. Isakov, Zhihui Wang, David Wecker, Daniel A. Lidar, John M. Martinis, and Matthias Troyer. Evidence for quantum annealing with more than one hundred qubits. *Nature Physics*, 10(3):218–224, March 2014. Number: 3 Publisher: Nature Publishing Group.
- [9] William Fuller Brown. Thermal Fluctuations of a Single-Domain Particle. *Physical Review*, 130(5):1677–1686, June 1963.
- [10] Tingsu Chen. Integration of GMR-based spin torque oscillators and CMOS circuitry. page 9, 2015.

- [11] Tingsu Chen, Randy K. Dumas, Anders Eklund, Pranaba K. Muduli, Afshin Houshang, Ahmad A. Awad, Philipp Dürrenfeld, B. Gunnar Malm, Ana Rusu, and Johan Åkerman. Spin-Torque and Spin-Hall Nano-Oscillators. *Proceedings of the IEEE*, 104(10):1919–1945, October 2016. Conference Name: Proceedings of the IEEE.
- [12] Tingsu Chen, Anders Eklund, Ezio Iacocca, Saul Rodriguez, B. Gunnar Malm, Johan Åkerman, and Ana Rusu. Comprehensive and Macrospin-Based Magnetic Tunnel Junction Spin Torque Oscillator Model- Part II: Verilog-A Model Implementation. *IEEE Transactions on Electron Devices*, 62(3):1045–1051, March 2015. Conference Name: IEEE Transactions on Electron Devices.
- [13] Francesco Chicco, Alessandro Pezzotta, and Christian C. Enz. Analysis of power consumption in LC oscillators based on the inversion coefficient. In *2017 IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 1–4, Baltimore, MD, USA, May 2017. IEEE.
- [14] Jeffrey Chou, Suraj Bramhavar, Siddhartha Ghosh, and William Herzog. Analog Coupled Oscillator Based Weighted Ising Machine. *Scientific Reports*, 9(1):14786, October 2019. Number: 1 Publisher: Nature Publishing Group.
- [15] E. Cohen and B. Tamir. Quantum annealing – foundations and frontiers. *The European Physical Journal Special Topics*, 224(1):89–110, February 2015.
- [16] William Cruz-Santos, Salvador E. Venegas-Andraca, and Marco Lanzagorta. A QUBO Formulation of Minimum Multicut Problem Instances in Trees for D-Wave Quantum Annealers. *Scientific Reports*, 9(1):17216, November 2019. Number: 1 Publisher: Nature Publishing Group.
- [17] György Csaba, Matt Pufall, William Rippard, and Wolfgang Porod. Modeling of coupled spin torque oscillators for applications in associative memories. In *2012 12th IEEE International Conference on Nanotechnology (IEEE-NANO)*, pages 1–4, August 2012. ISSN: 1944-9399.
- [18] A. Demir, A. Mehrotra, and J. Roychowdhury. Phase noise in oscillators: a unifying theory and numerical methods for characterization. *IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications*, 47(5):655–674, May 2000. Conference Name: IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications.
- [19] Alper Demir. Floquet theory and non-linear perturbation analysis for oscillators with differential-algebraic equations. *International Journal of Circuit Theory and Applications*, 28(2):163–185, 2000. _eprint: [https://onlinelibrary.wiley.com/doi/pdf/10.1002/%28SICI%291097-007X%28200003/04%2928%3A2%3C163%3A%3AAID-CTA101%3E3.0.CO%3B2-K](https://onlinelibrary.wiley.com/doi/pdf/10.1002/%28SICI%291097-007X%28200003%2F04%2928%3A2%3C163%3A%3AAID-CTA101%3E3.0.CO%3B2-K).

- [20] S. Dutta, A. Khanna, and S. Datta. Understanding the Continuous-Time Dynamics of Phase-Transition Nano-Oscillator-Based Ising Hamiltonian Solver. *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, 6(2):155–163, December 2020. Conference Name: IEEE Journal on Exploratory Solid-State Computational Devices and Circuits.
- [21] Sourav Dutta, Abhishek Khanna, Hanjong Paik, Darrell Schlom, Arijit Raychowdhury, Zoltan Toroczkai, and Suman Datta. An Ising Hamiltonian Solver using Stochastic Phase-Transition Nano-Oscillators. preprint, In Review, November 2020.
- [22] Sourav Dutta, Abhinav Parihar, Abhishek Khanna, Jorge Gomez, Wriddhi Chakraborty, Matthew Jerry, Benjamin Grisafe, Arijit Raychowdhury, and Suman Datta. Programmable coupled oscillators for synchronized locomotion. *Nature Communications*, 10(1):3299, July 2019. Number: 1 Publisher: Nature Publishing Group.
- [23] A. M. Ferreiro, J. A. García, J. G. López-Salas, and C. Vázquez. An efficient implementation of parallel simulated annealing algorithm in GPUs. *Journal of Global Optimization*, 57(3):863–890, November 2013.
- [24] T.L. Gilbert. A phenomenological theory of damping in ferromagnetic materials. *IEEE Transactions on Magnetism*, 40(6):3443–3449, November 2004. Conference Name: IEEE Transactions on Magnetism.
- [25] B. Gu, K.K. Gullapalli, S. Hamm, B. Mulvaney, X. Lai, and J. Roychowdhury. Implementing nonlinear oscillator macromodels using Verilog-AMS for accurate prediction of injection locking behaviors of oscillators. In *BMAS 2005. Proceedings of the 2005 IEEE International Behavioral Modeling and Simulation Workshop, 2005.*, pages 43–47, September 2005. ISSN: 2160-3812.
- [26] A. Hajimiri and T.H. Lee. A general theory of phase noise in electrical oscillators. *IEEE Journal of Solid-State Circuits*, 33(2):179–194, February 1998. Conference Name: IEEE Journal of Solid-State Circuits.
- [27] Ryan Hamerly, Takahiro Inagaki, Peter L. McMahon, Davide Venturelli, Alireza Marandi, Tatsuhiro Onodera, Edwin Ng, Carsten Langrock, Kensuke Inaba, Toshi-mori Honjo, Koji Enbutsu, Takeshi Umeki, Ryoichi Kasahara, Shoko Utsunomiya, Satoshi Kako, Ken-ichi Kawarabayashi, Robert L. Byer, Martin M. Fejer, Hideo Mabuchi, Dirk Englund, Eleanor Rieffel, Hiroki Takesue, and Yoshihisa Yamamoto. Experimental investigation of performance differences between coherent Ising machines and a quantum annealer. *Science Advances*, 5(5):eaau0823, May 2019. Publisher: American Association for the Advancement of Science Section: Research Article.
- [28] Afshin Houshang, Mohammad Zahedinejad, Shreyas Muralidhar, Jakub Chęcinski, Ahmad A. Awad, and Johan Åkerman. A Spin Hall Ising Machine. *arXiv:2006.02236 [cond-mat, physics:physics]*, June 2020. arXiv: 2006.02236.

- [29] T. Inagaki, Y. Haribara, K. Igarashi, T. Sonobe, S. Tamate, T. Honjo, A. Marandi, P. L. McMahon, T. Umeki, K. Enbutsu, O. Tadanaga, H. Takenouchi, K. Aihara, K.-i. Kawarabayashi, K. Inoue, S. Utsunomiya, and H. Takesue. A coherent Ising machine for 2000-node optimization problems. *Science*, 354(6312):603–606, November 2016.
- [30] L. Ingber. Simulated annealing: Practice versus theory. *Mathematical and Computer Modelling*, 18(11):29–57, December 1993.
- [31] Ernst Ising. Beitrag zur Theorie des Ferromagnetismus. *Zeitschrift für Physik*, 31(1):253–258, February 1925.
- [32] Mark Johnson, Mohammad Amin, S Gildert, Trevor Lanting, F Hamze, N Dickson, R Harris, Andrew Berkley, Jan Johansson, Paul Bunyk, E Chapple, C Enderud, Jeremy Hilton, Kamran Karimi, E Ladizinsky, Nicolas Ladizinsky, T Oh, I Perminov, C Rich, and Geordie Rose. Quantum annealing with manufactured spins. *Nature*, 473:194–8, May 2011.
- [33] R. M. Karp. On the Computational Complexity of Combinatorial Problems. *Networks*, 5(1):45–68, 1975. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/net.1975.5.1.45>.
- [34] Mohammad Kazemi, Graham E. Rowlands, Engin Ipek, Robert A. Buhrman, and Eby G. Friedman. Compact Model for Spin–Orbit Magnetic Tunnel Junctions. *IEEE Transactions on Electron Devices*, 63(2):848–855, February 2016. Conference Name: IEEE Transactions on Electron Devices.
- [35] Guru Khalsa, M. D. Stiles, and J. Grollier. Critical current and linewidth reduction in spin-torque nano-oscillators by delayed self-injection. *Applied Physics Letters*, 106(24):242402, June 2015. Publisher: American Institute of Physics.
- [36] Joo-Von Kim, Vasil Tiberkevich, and Andrei N Slavin. Generation Linewidth of an Auto-Oscillator with a Nonlinear Frequency Shift: Spin-Torque Nano-Oscillator. *PHYSICAL REVIEW LETTERS*, page 4, 2008.
- [37] Andrew D. King, William Bernoudy, James King, Andrew J. Berkley, and Trevor Lanting. Emulating the coherent Ising machine with a mean-field algorithm. *arXiv:1806.08422 [quant-ph]*, June 2018. arXiv: 1806.08422.
- [38] S. I. Kiselev, J. C. Sankey, I. N. Krivorotov, N. C. Emley, R. J. Schoelkopf, R. A. Buhrman, and D. C. Ralph. Microwave oscillations of a nanomagnet driven by a spin-polarized current. *Nature*, 425(6956):380–383, September 2003. Number: 6956 Publisher: Nature Publishing Group.
- [39] Kiwamu Kudo, Tazumi Nagasawa, Koichi Mizushima, Hirofumi Suto, and Rie Sato. Numerical Simulation on Temporal Response of Spin-Torque Oscillator to Magnetic Pulses. *Appl. Phys. Express*, page 4, 2010.

- [40] Kiwamu Kudo, Tazumi Nagasawa, Rie Sato, and Koichi Mizushima. Amplitude-phase coupling in a spin-torque nano-oscillator. *Journal of Applied Physics*, 105(7):07D105, February 2009. Publisher: American Institute of Physics.
- [41] Yoshiki Kuramoto. Cooperative Dynamics of Oscillator Community: A Study Based on Lattice of Rings. *Progress of Theoretical Physics Supplement*, 79:223–240, 1984.
- [42] Werner Köster. *Beiträge zur Theorie des Ferromagnetismus und der Magnetisierungskurve*. Springer-Verlag, July 2013. Google-Books-ID: rQ7NBgAAQBAJ.
- [43] Xiaolue Lai and J. Roychowdhury. Capturing oscillator injection locking via nonlinear phase-domain macromodels. *IEEE Transactions on Microwave Theory and Techniques*, 52(9):2251–2261, September 2004. Conference Name: IEEE Transactions on Microwave Theory and Techniques.
- [44] L Landau and E Lifshits. ON THE THEORY OF THE DISPERSION OF MAGNETIC PERMEABILITY IN FERROMAGNETIC BODIES. 53:9.
- [45] R. Lebrun, A. Jenkins, A. Dussaux, N. Locatelli, S. Tsunegi, E. Grimaldi, H. Kubota, P. Bortolotti, K. Yakushiji, J. Grollier, A. Fukushima, S. Yuasa, and V. Cros. Understanding of Phase Noise Squeezing Under Fractional Synchronization of a Nonlinear Spin Transfer Vortex Oscillator. *Physical Review Letters*, 115(1):017201, June 2015.
- [46] Luqiao Liu, O. J. Lee, T. J. Gudmundsen, D. C. Ralph, and R. A. Buhrman. Current-Induced Switching of Perpendicularly Magnetized Magnetic Layers Using Spin Torque from the Spin Hall Effect. *Physical Review Letters*, 109(9):096602, August 2012.
- [47] Luqiao Liu, Chi-Feng Pai, D. C. Ralph, and R. A. Buhrman. Magnetic Oscillations Driven by the Spin Hall Effect in 3-Terminal Magnetic Tunnel Junction Devices. *Physical Review Letters*, 109(18):186602, October 2012.
- [48] Andrew Lucas. Ising formulations of many NP problems. *Frontiers in Physics*, 2, 2014. Publisher: Frontiers.
- [49] P. Maffezzoni and D. D’Amore. Phase-Noise Reduction in Oscillators via Small-Signal Injection. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 58(10):2498–2507, October 2011. Conference Name: IEEE Transactions on Circuits and Systems I: Regular Papers.
- [50] Paolo Maffezzoni. Synchronization Analysis of Two Weakly Coupled Oscillators Through a PPV Macromodel. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 57(3):654–663, March 2010. Conference Name: IEEE Transactions on Circuits and Systems I: Regular Papers.

- [51] Alireza Marandi, Zhe Wang, Kenta Takata, Robert L. Byer, and Yoshihisa Yamamoto. Network of time-multiplexed optical parametric oscillators as a coherent Ising machine. *Nature Photonics*, 8(12):937–942, December 2014.
- [52] T. Maruyama, Y. Shiota, T. Nozaki, K. Ohta, N. Toda, M. Mizuguchi, A. A. Tulapurkar, T. Shinjo, M. Shiraishi, S. Mizukami, Y. Ando, and Y. Suzuki. Large voltage-induced magnetic anisotropy change in a few atomic layers of iron. *Nature Nanotechnology*, 4(3):158–161, March 2009. Number: 3 Publisher: Nature Publishing Group.
- [53] Peter L McMahon, Alireza Marandi, Yoshitaka Haribara, Ryan Hamerly, Carsten Langrock, Shuhei Tamate, Takahiro Inagaki, Hiroki Takesue, Shoko Utsunomiya, Kazuyuki Aihara, Robert L Byer, M M Fejer, Hideo Mabuchi, and Yoshihisa Yamamoto. A fully programmable 100-spin coherent Ising machine with all-to-all connections. page 5.
- [54] Georgios D. Panagopoulos, Charles Augustine, and Kaushik Roy. Physics-Based SPICE-Compatible Compact Model for Simulating Hybrid MTJ/CMOS Circuits. *IEEE Transactions on Electron Devices*, 60(9):2808–2814, September 2013. Conference Name: IEEE Transactions on Electron Devices.
- [55] Federico Pepe, Andrea Bonfanti, Salvatore Levantino, Paolo Maffezzoni, Carlo Samori, and Andrea L. Lacaita. An efficient linear-time variant simulation technique of oscillator phase sensitivity function. In *2012 International Conference on Synthesis, Modeling, Analysis and Simulation Methods and Applications to Circuit Design (SMACD)*, pages 17–20, September 2012.
- [56] Bogdan Popescu, Gyorgy Csaba, Dan Popescu, Amir Hossein Fallahpour, Paolo Lugli, Wolfgang Porod, and Markus Becherer. Simulation of coupled spin torque oscillators for pattern recognition. *Journal of Applied Physics*, 124(15):152128, October 2018. Publisher: American Institute of Physics.
- [57] Miguel Romera, Philippe Talatchian, Sumito Tsunegi, Flavio Abreu Araujo, Vincent Cros, Paolo Bortolotti, Juan Trastoy, Kay Yakushiji, Akio Fukushima, Hitoshi Kubota, Shinji Yuasa, Maxence Ernoult, Damir Vodenicarevic, Tifenn Hirtzlin, Nicolas Locatelli, Damien Querlioz, and Julie Grollier. Vowel recognition with four coupled spin-torque nano-oscillators. *Nature*, 563(7730):230–234, November 2018. Number: 7730 Publisher: Nature Publishing Group.
- [58] Jaijeet Roychowdhury. Multi-Time PDEs for Dynamical System Analysis. pages 85–143. July 2011.
- [59] Shigeru Shinomoto and Y. Oshiki Kuramoto. Phase transitions in active rotator systems. *Prog. Theor. Phys*, 1986.
- [60] A. N. Slavin and V. S. Tiberkevich. Nonlinear self-phase-locking effect in an array of current-driven magnetic nanocontacts. *Physical Review B*, 72(9):092407, September 2005.

- [61] Andrei Slavin and Vasil Tiberkevich. Nonlinear Auto-Oscillator Theory of Microwave Generation by Spin-Polarized Current. *IEEE Transactions on Magnetics*, 45(4):1875–1918, April 2009. Conference Name: IEEE Transactions on Magnetics.
- [62] J.C. Slonczewski. Current-driven excitation of magnetic multilayers. *Journal of Magnetism and Magnetic Materials*, 159(1-2):L1–L7, June 1996.
- [63] Kate A Smith. Neural Networks for Combinatorial Optimization: A Review of More Than a Decade of Research. page 21.
- [64] Tomohiro Taniguchi, Hiroko Arai, Sumito Tsunegi, Shingo Tamaru, Hitoshi Kubota, and Hiroshi Imamura. Critical Field of Spin Torque Oscillator with Perpendicularly Magnetized Free Layer. *Applied Physics Express*, 6(12):123003, December 2013.
- [65] Tomohiro Taniguchi, Yasuhiro Utsumi, Michael Marthaler, Dmitri S. Golubev, and Hiroshi Imamura. Spin torque switching of an in-plane magnetized system in a thermally activated region. *Physical Review B*, 87(5):054406, February 2013.
- [66] M. Tarequzzaman, T. Böhnert, M. Decker, J. D. Costa, J. Borme, B. Lacoste, E. Paz, A. S. Jenkins, S. Serrano-Guisan, C. H. Back, R. Ferreira, and P. P. Freitas. Spin torque nano-oscillator driven by combined spin injection from tunneling and spin Hall current. *Communications Physics*, 2(1):1–8, February 2019. Number: 1 Publisher: Nature Publishing Group.
- [67] Vasil Tiberkevich, Andrei Slavin, Elena Bankowski, and Grant Gerhart. Phase-locking and frustration in an array of nonlinear spin-torque nano-oscillators. *Applied Physics Letters*, 95(26):262505, December 2009. Publisher: American Institute of Physics.
- [68] Vasil S. Tiberkevich, Roman S. Khymyn, Hong X. Tang, and Andrei N. Slavin. Sensitivity to external signals and synchronization properties of a non-isochronous auto-oscillator with delayed feedback. *Scientific Reports*, 4(1):3873, January 2014. Number: 1 Publisher: Nature Publishing Group.
- [69] Jacob Torrejon, Mathieu Riou, Flavio Abreu Araujo, Sumito Tsunegi, Guru Khalsa, Damien Querlioz, Paolo Bortolotti, Vincent Cros, Kay Yakushiji, Akio Fukushima, Hitoshi Kubota, Shinji Yuasa, Mark D. Stiles, and Julie Grollier. Neuromorphic computing with nanoscale spintronic oscillators. *Nature*, 547(7664):428–431, July 2017. Number: 7664 Publisher: Nature Publishing Group.
- [70] Sumito Tsunegi, Eva Grimaldi, Romain Lebrun, Hitoshi Kubota, Alex S. Jenkins, Kay Yakushiji, Akio Fukushima, Paolo Bortolotti, Julie Grollier, Shinji Yuasa, and Vincent Cros. Self-Injection Locking of a Vortex Spin Torque Oscillator by Delayed Feedback. *Scientific Reports*, 6(1):26849, May 2016. Number: 1 Publisher: Nature Publishing Group.

- [71] Sergei Urazhdin, Phillip Tabor, Vasil Tiberkevich, and Andrei Slavin. Fractional Synchronization of Spin-Torque Nano-Oscillators. *Physical Review Letters*, 105(10):104101, August 2010.
- [72] Peter J. M. van Laarhoven and Emile H. L. Aarts. *Simulated Annealing: Theory and Applications*. Springer Netherlands, Dordrecht, 1987.
- [73] Arne Vansteenkiste, Jonathan Leliaert, Mykola Dvornik, Mathias Helsen, Felipe Garcia-Sanchez, and Bartel Van Waeyenberge. The design and verification of MuMax3. *AIP Advances*, 4(10):107133, October 2014. Publisher: American Institute of Physics.
- [74] Damir Vodenicarevic, Nicolas Locatelli, Flavio Abreu Araujo, Julie Grollier, and Damien Querlioz. A Nanotechnology-Ready Computing Scheme based on a Weakly Coupled Oscillator Network. *Scientific Reports*, 7(1):44772, March 2017. Number: 1 Publisher: Nature Publishing Group.
- [75] Tianshi Wang and Jaijeet Roychowdhury. OIM: Oscillator-Based Ising Machines for Solving Combinatorial Optimisation Problems. In Ian McQuillan and Shinosuke Seki, editors, *Unconventional Computation and Natural Computation*, volume 11493, pages 232–256. Springer International Publishing, Cham, 2019. Series Title: Lecture Notes in Computer Science.
- [76] Tianshi Wang, Leon Wu, and Jaijeet Roychowdhury. Late Breaking Results: New Computational Results and Hardware Prototypes for Oscillator-based Ising Machines. In *2019 56th ACM/IEEE Design Automation Conference (DAC)*, pages 1–2, June 2019. ISSN: 0738-100X.
- [77] Jiang Xiao, A. Zangwill, and M. D. Stiles. Macrospin models of spin transfer dynamics. *Physical Review B*, 72(1):014446, July 2005.
- [78] Xiaolue Lai and J. Roychowdhury. Fast and accurate simulation of coupled oscillators using nonlinear phase macromodels. In *IEEE MTT-S International Microwave Symposium Digest, 2005.*, pages 4 pp.–, June 2005. ISSN: 0149-645X.
- [79] Mihalis Yannakakis. Node-and edge-deletion NP-complete problems. In *Proceedings of the tenth annual ACM symposium on Theory of computing - STOC '78*, pages 253–264, San Diego, California, United States, 1978. ACM Press.
- [80] Karthik Yogendra, Deliang Fan, and Kaushik Roy. Coupled Spin Torque Nano Oscillators for Low Power Neural Computation. *IEEE Transactions on Magnetism*, 51(10):1–9, October 2015. Conference Name: IEEE Transactions on Magnetism.
- [81] Chihiro Yoshimura, Masato Hayashi, Takuya Okuyama, and Masanao Yamaoka. FPGA-based Annealing Processor for Ising Model. In *2016 Fourth International Symposium on Computing and Networking (CANDAR)*, pages 436–442, November 2016. ISSN: 2379-1896.

- [82] S Yuasa and D D Djayaprawira. Giant tunnel magnetoresistance in magnetic tunnel junctions with a crystalline MgO(0 0 1) barrier. *Journal of Physics D: Applied Physics*, 40(21):R337–R354, November 2007.
- [83] Lang Zeng, Zuodong Zhang, Haoxuan Chen, Tianqi Gao, Deming Zhang, Mingzhi Long, Youguang Zhang, and Weisheng Zhao. Modulation and Demodulation of Digital Frequency Shift Keying System Based on Spin Torque Nano Oscillator with Voltage Controlled Magnetic Anisotropy Effect. In *2019 IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 1–5, May 2019. ISSN: 2158-1525.